



DIFFER

Dutch Institute for
Fundamental Energy Research

XARMAE Workshop, Barcelona, 8-10 January 2014

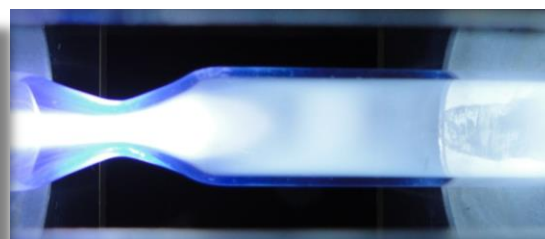
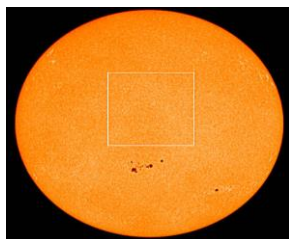
Chemical Energy storage by CO₂ Plasmolysis

by Adelbert Goede¹, W.A.Bongers¹, M.F.Graswinckel¹, M.C.M van de Sanden¹, M.Leins², J.Kopecki², A.Schulz², M.Walker²

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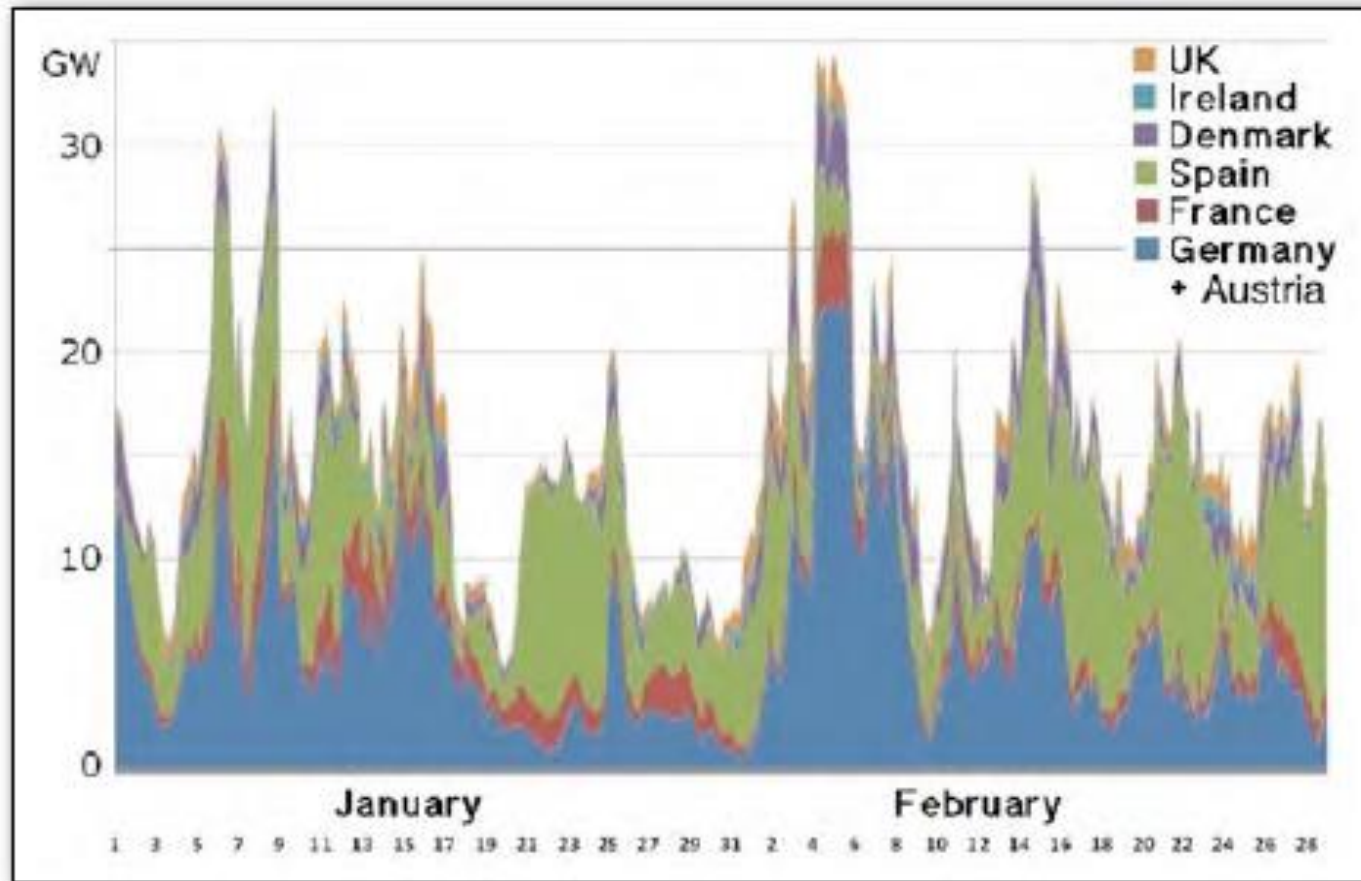


Why Chemical Storage

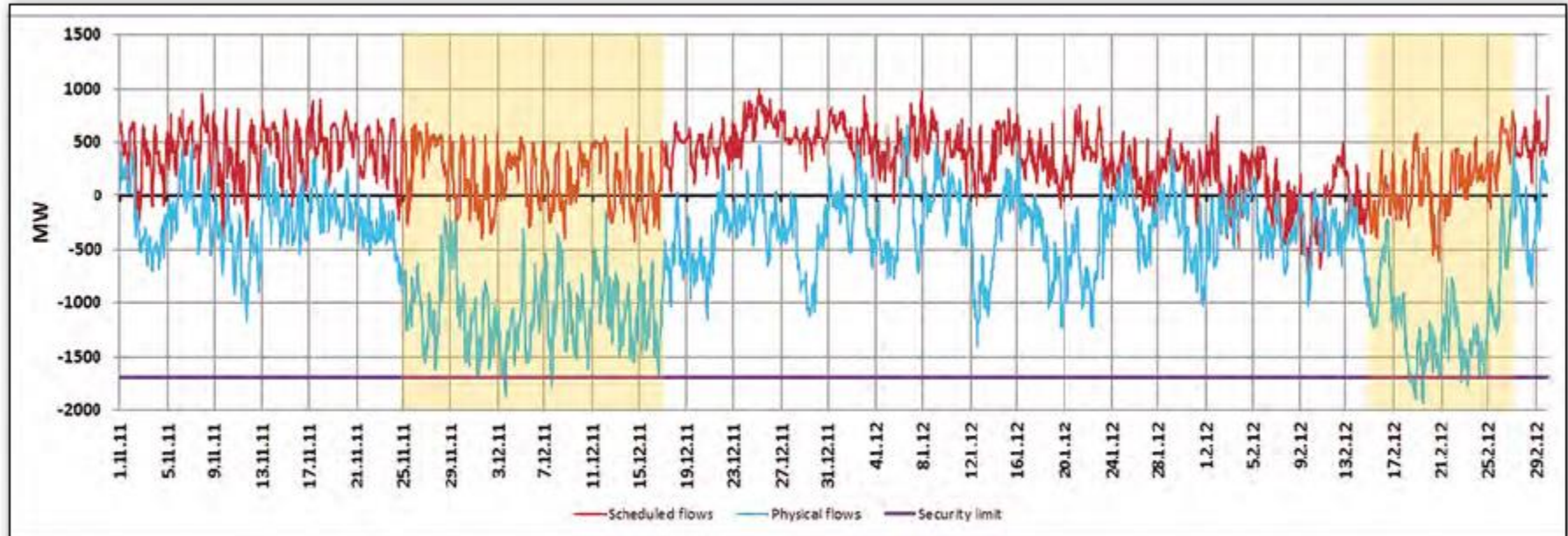
- **Renewable Energy Sources** (PV, wind, concentrated solar) are limited by ill-matched Supply-Demand character
 - ◆ Intermittent peak power flow requires over capacity on grid
 - ◆ Installed capacity to provide base load is vastly under-employed
- **Storage** capacity to bridge one day EU electrical power requirement $\sim 10\text{TWh}$. EU hydro storage capacity 15TWh .
- EU storage requirements over several days can only be met by **chemical storage**

Which Energy policy for France 2010-2050

Jean Louis Bobin, EuroPhysNews 44/1 2013



Wind power over Western Europe Jan-Feb 2011 by H.Flocard, J.P.Pervès)
Peaks and troughs are correlated due to size of low pressure areas
Installed wind capacity Europe 96 GW (2011)



Power (MW) vs. time period Nov 2011 to Feb 2012

- red: ČEPS scheduled flow (~ 500MW generated)
- blue: Physical flow (German RES import)
- violet: Max safe operation limit 1700MW exceeded

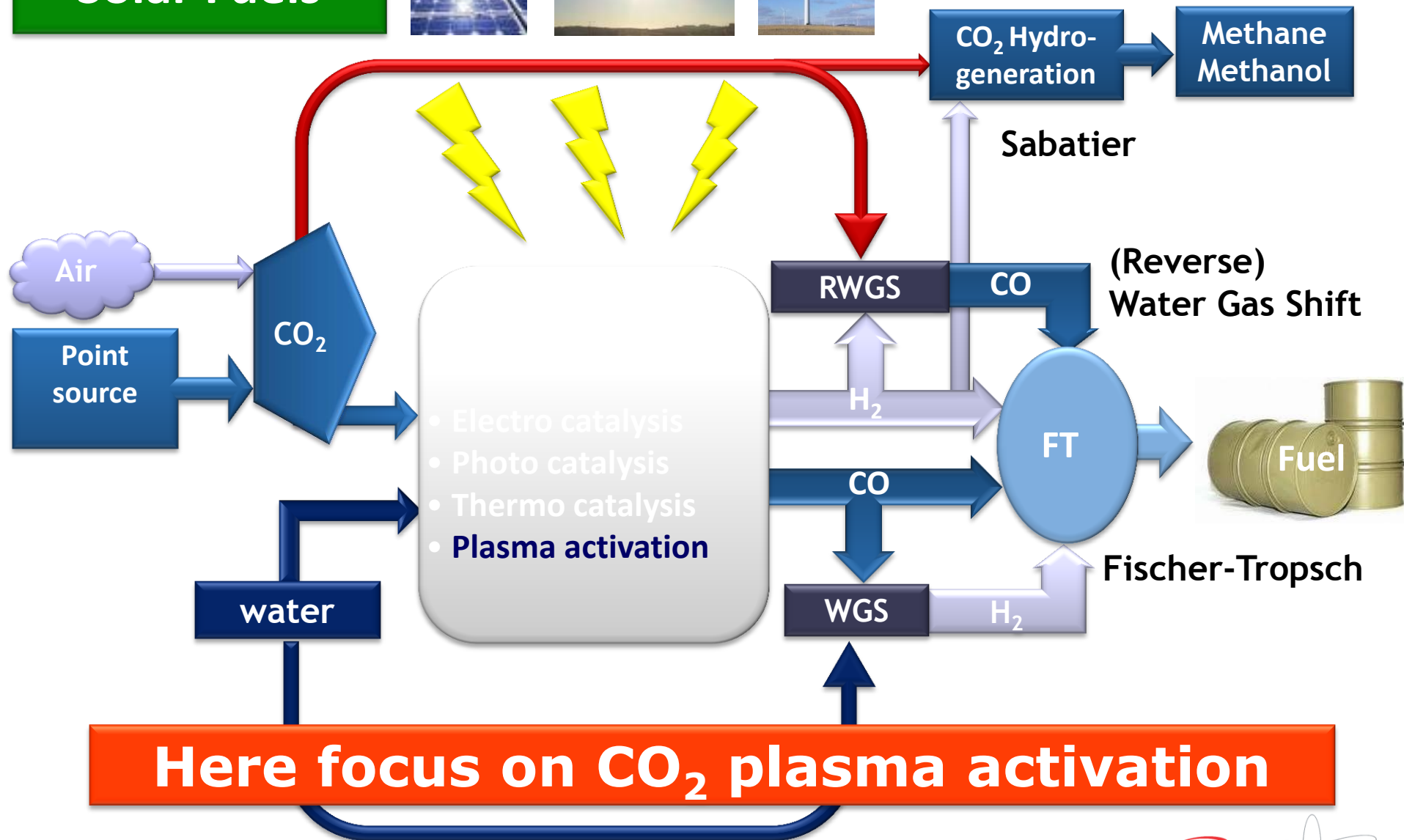


Power to Gas & Carbon Capture and Usage

One way to cope with intermittent supply
and to enhance useful capacity

- Power to Gas (**P2G**) to convert excess **electricity** generated by wind/solar power **into hydro-carbon-based (solar) fuels** by splitting CO_2 and or H_2O and forming syngas
- CO_2 neutral power generation possible when **CO_2** emitted by burning these fuels is **recaptured and reused (CCU)**
- Advantage: **Existing infra structure** for gas and oil storage, transport and distribution can be employed and enhanced

Grand Scheme Solar Fuels





Why Plasma?

Low temperature ($T_e \sim 1\text{eV}$, $T_i \sim 0.1\text{eV}$) weakly ionised (10^{-5} - 10^{-6}) gas to ease conditions for splitting CO_2 by vibrational excitation of CO_2 , which lowers activation energy.

Conditions far from thermo-dynamic equilibrium $T_{\text{vib}} \gg T_{\text{gas}}$

Compared with electrolysis

- High energy efficiency ($\sim 60\%$ demonstrated)
- High power density ($45\text{W}/\text{cm}^3$, CO $20\text{A}/\text{cm}^2$ equivalent)
- Rapid ramping up and down (wrt high temperature SOEC)
- No scarce materials employed (Pt catalyst in PEM)



Out of equilibrium $T_{\text{vib}} > T_0$ chemistry

Chemical reaction scheme



followed by reuse energetic O radical

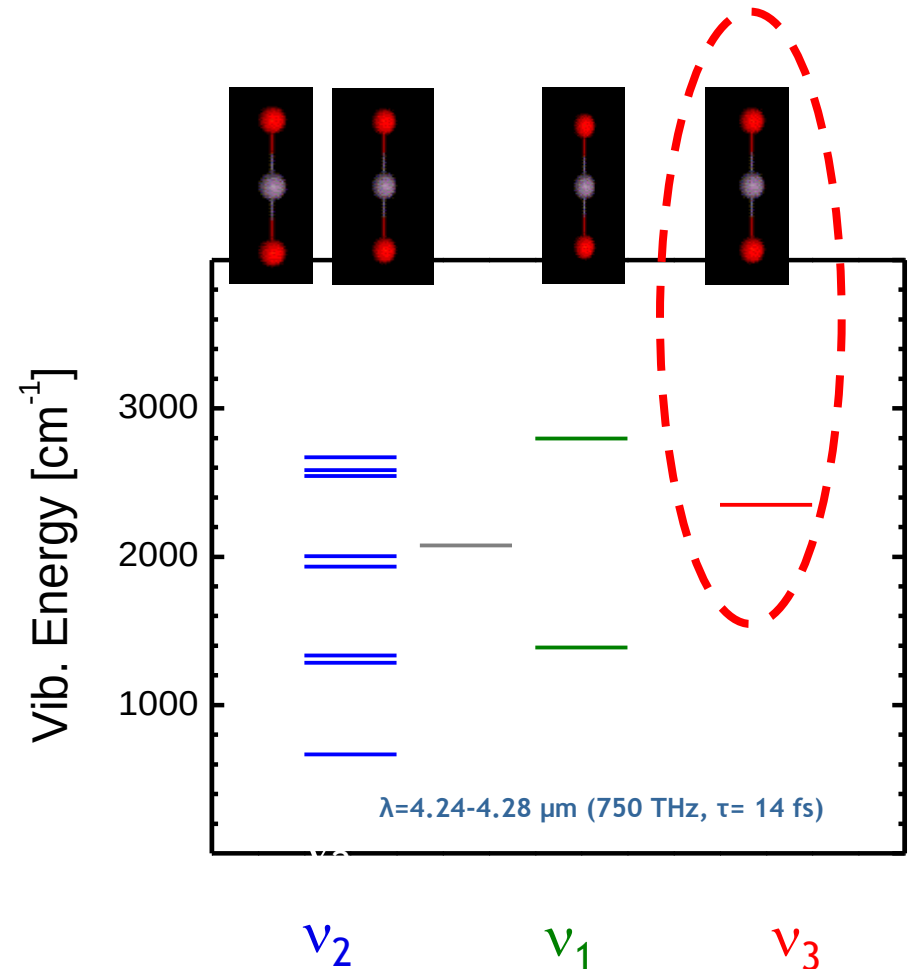


Net



Efficiency to be increased by

Concentration of electron energy
on vibrational excitation of CO_2 in
asymmetric stretch mode

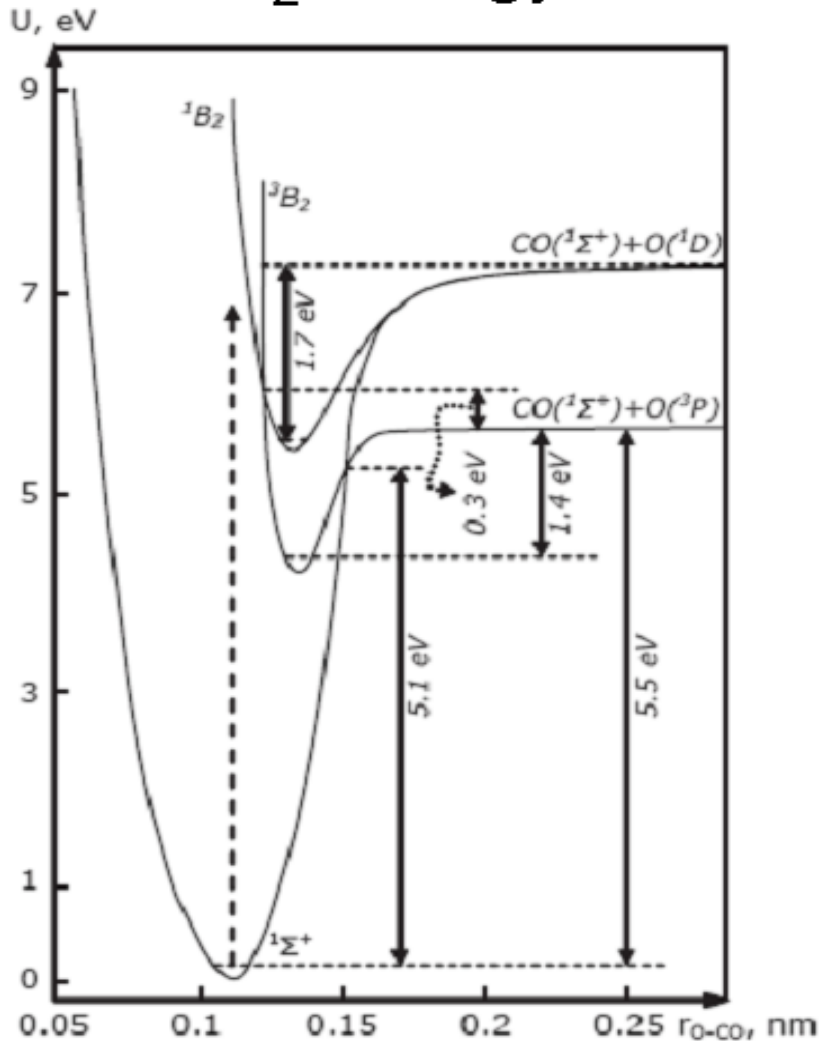


Arrhenius/Fridman: Activation energy reduced by vibration energy
 $k = A \exp (\alpha E_{\text{v}} - E_{\text{a}}) / kT$



CO₂ chemistry out of equilibrium $T_v > T_0$

low CO₂ energy levels



- Direct dissociation via $1B_2$ vibration mode produces singlet O¹D atomic oxygen requiring ~7.5 eV
- More efficient via $1B_2$ to $3B_2$ singlet triplet transition ~5.5 eV producing
- Atomic Oxygen triplet O³P



cross section CO₂ assym vibrational excitation

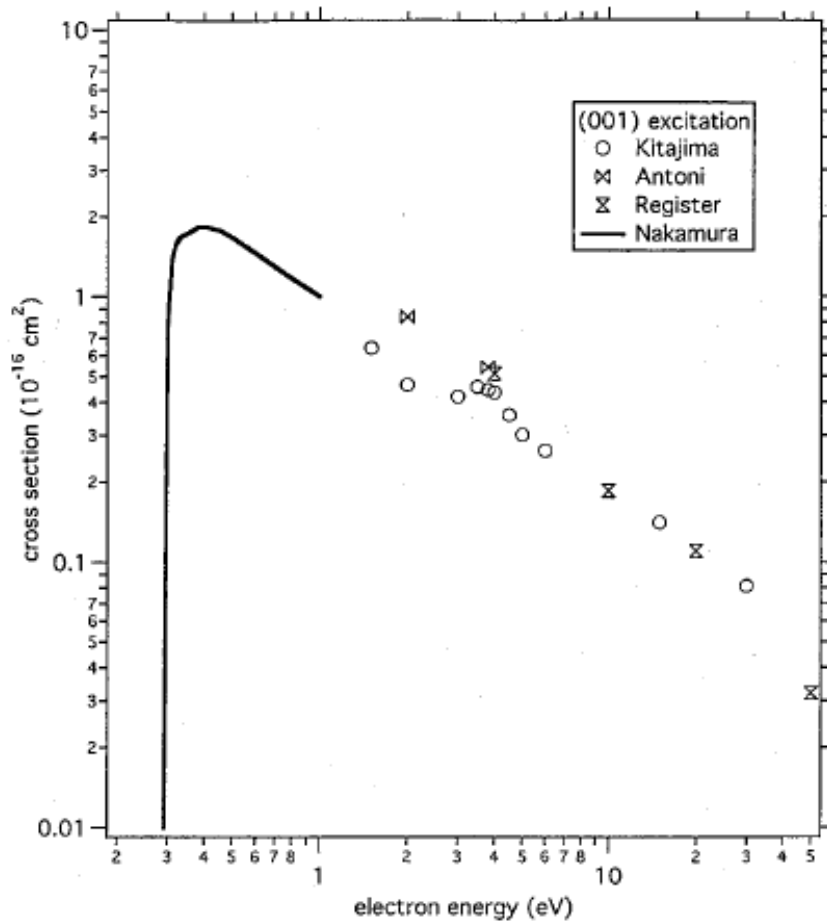
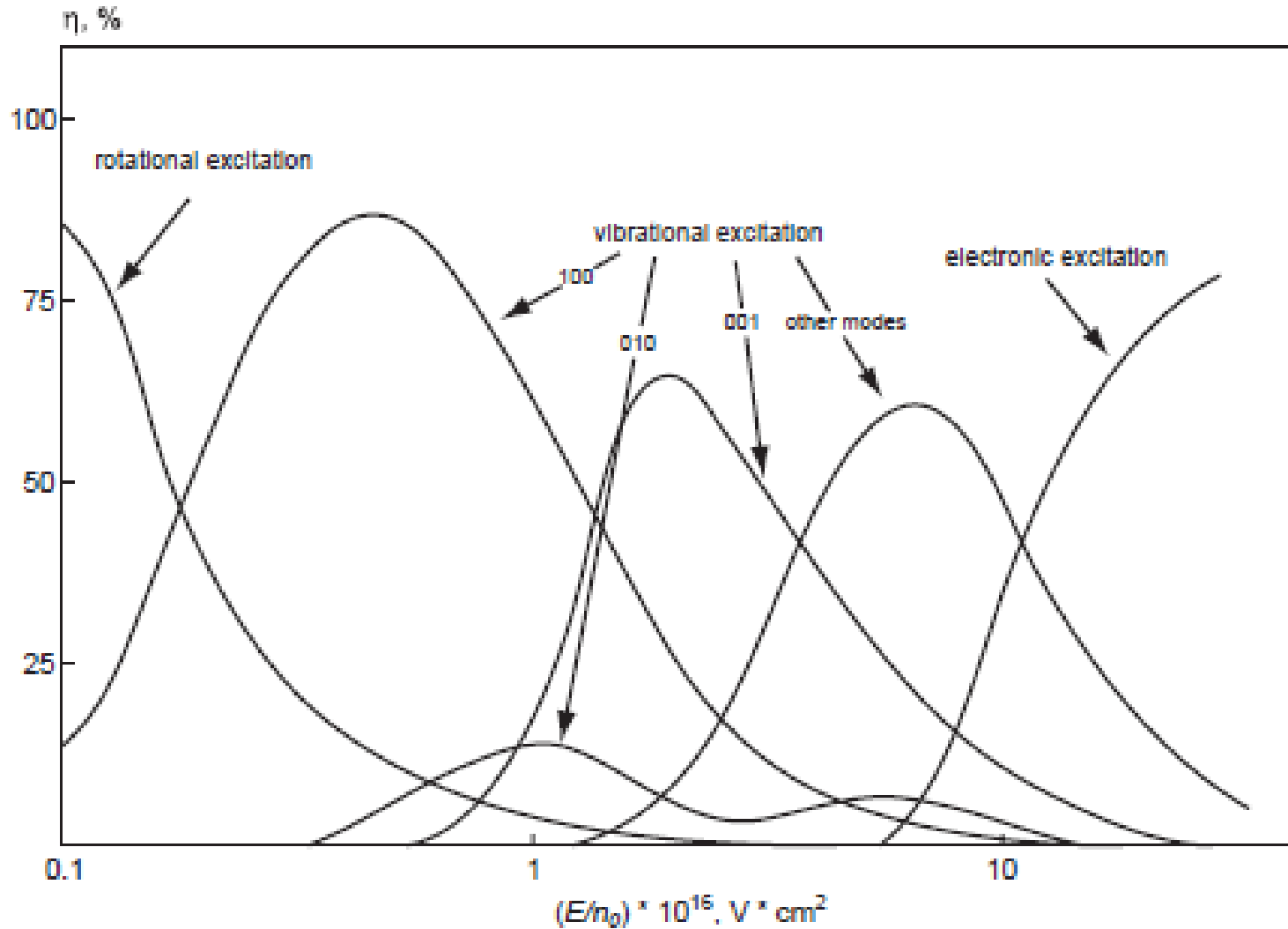


FIG. 7. Cross sections for the electron-impact excitation of the vibrational state (001) of CO₂. Comparison of the beam experiments by Kitajima *et al.*,³² Antoni *et al.*,³³ and Register *et al.*,²⁶ and the swarm result of Nakamura³¹ is shown.

- Vibration excitation of asymmetric stretch mode reaches maximum at electron energy 0.4 eV
- RF field at 915 MHz is stationary compared with collision processes ($\ll 1\text{ns}$)
- Steady electron drift establishes $v_D \sim E/n$
- $E/n\sigma = E\lambda$ equals potential drop an electron experiences in between collisions
- result $E/n = 1.4 \cdot 10^{-16} \text{Vcm}^2$



Vibrational excitation as function of reduced electric field E/n



ν_3 (001) asymmetric stretch mode @ $E/n = 2 \times 10^{-16} Vcm^2$



Energy Efficiency η against energy per incoming CO_2 molecule E_v

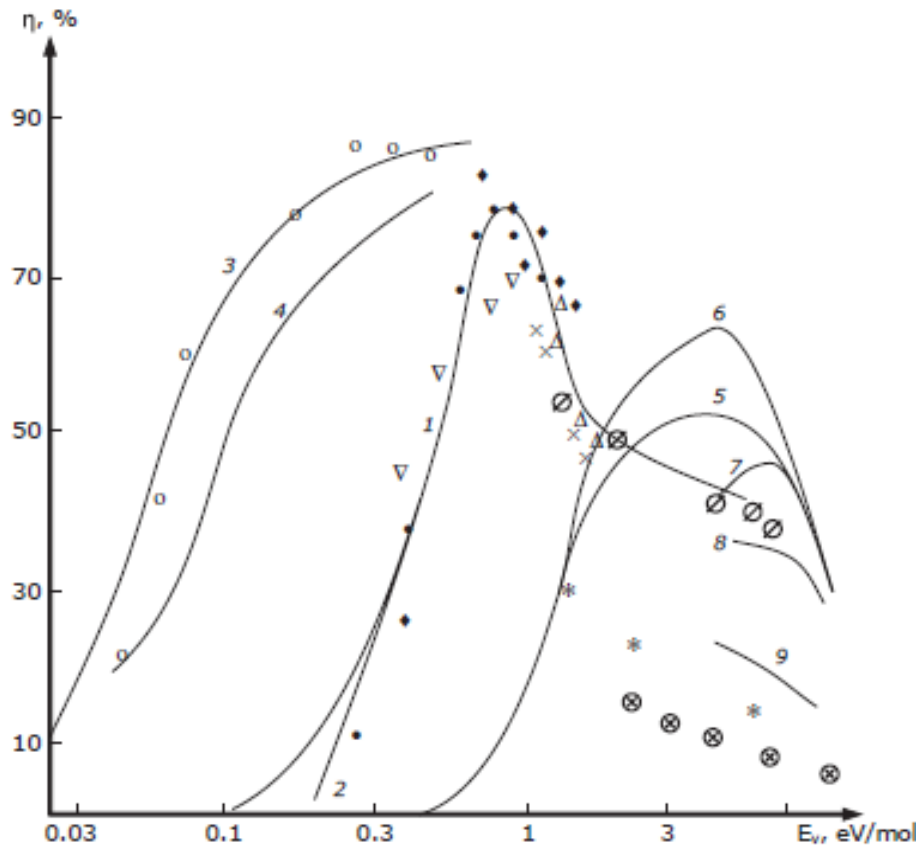


Figure 5-2. Energy efficiency of CO_2 dissociation as a function of specific energy input. (1, 2), non-equilibrium calculations in one- and two-approximations; non-equilibrium calculations for supersonic flows: (3) $M = 5$; (4) $M = 3.5$; calculations of thermal dissociation with (5) ideal and (6) super-ideal quenching; (7) thermal dissociation with quenching rates 10^9 K/s, (8) 10^8 K/s, (9) 10^7 K/s. Different experiments in microwave discharges: o, ♦, Δ, ×. Experiments in supersonic microwave discharges: ●. Experiments in different RF-CCP discharges: o, ∇. Experiments in RF-ICP discharges: ∅. Experiments in different arc discharges: ∅, *.

- $\eta = H/E_{\text{co}}$
- $H = 2.9$ eV net enthalpy to produce one CO molecule
- E_{co} energy per CO produced
- The fraction α of CO_2 converted into CO yields
- $E_v = \alpha E_{\text{co}} = (\alpha/\eta) H$
where E_v = energy per input CO_2 molecule



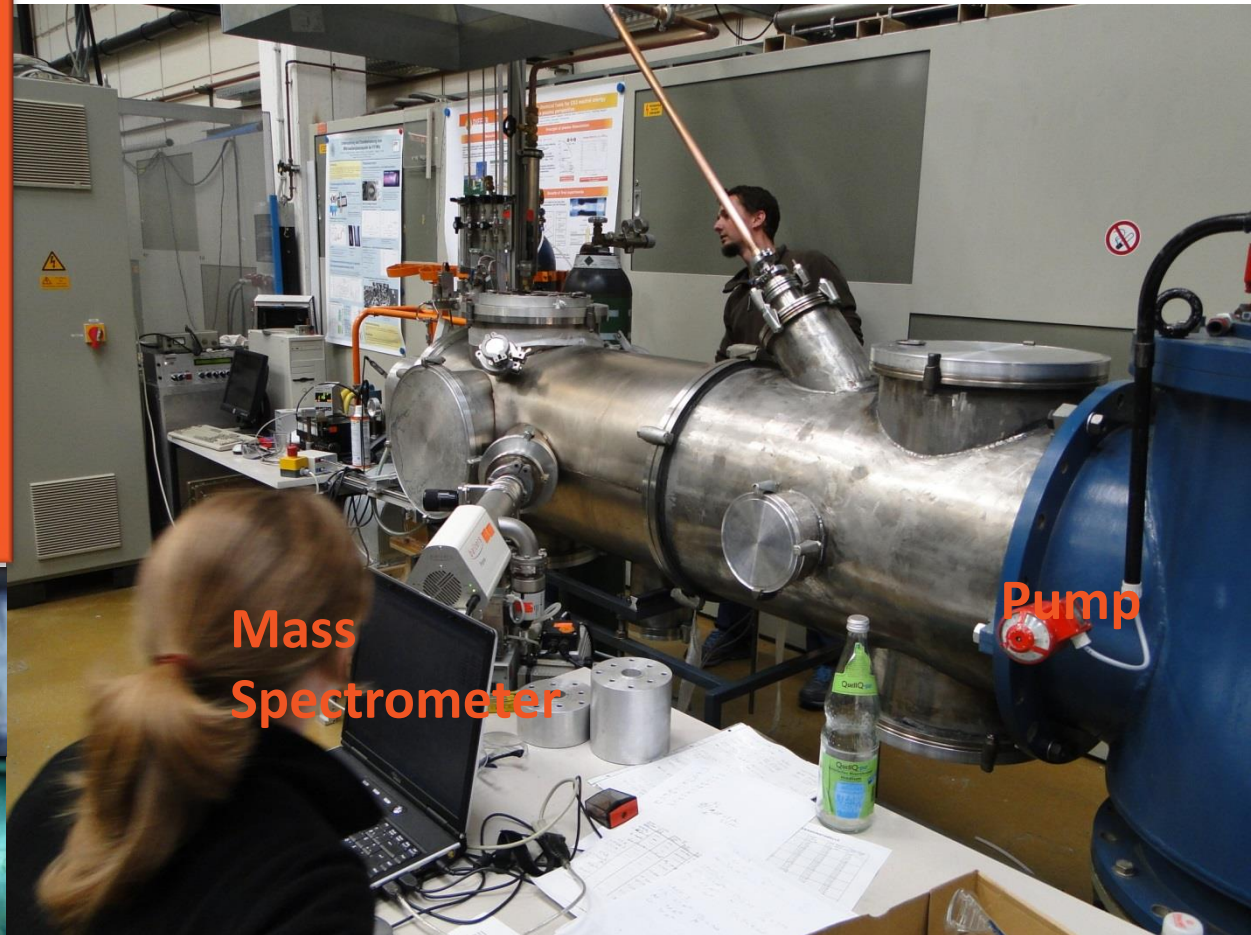
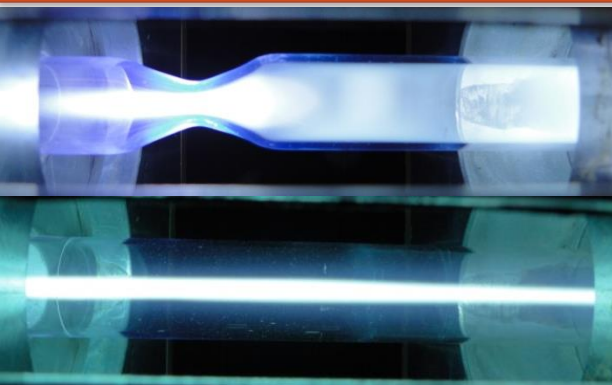
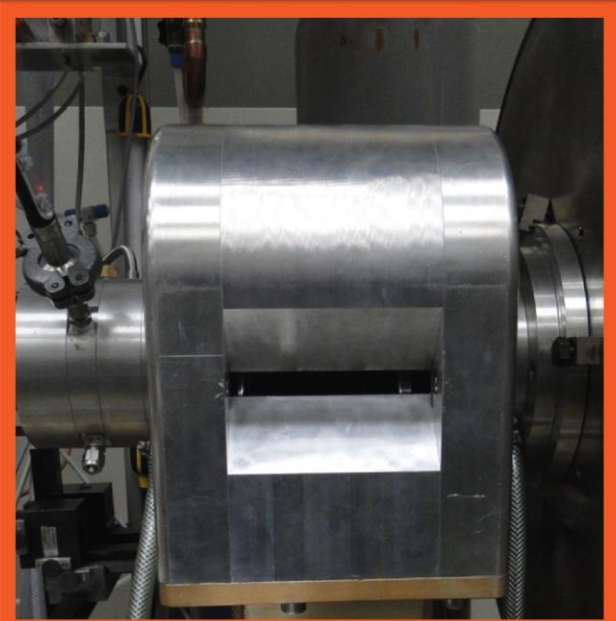
Particle and energy balance

- Particle balance $2v_B = n_0 \langle \sigma v \rangle r$ yields electron temperature through Bohm velocity $v_B = \sqrt{kT_e/m_i}$
- Energy balance $P_{RF} = n_e n_0 \sum_i \langle \sigma v \rangle \Delta \varepsilon_i V$ yields electron density and ionisation degree
- Typical ionisation degree $n_e/n_0 \sim 10^{-5} - 10^{-6}$
- Typical $kT_e \sim 1 - 3 \text{ eV}$,
- $T_e \sim E/n$ where $E/n \sim 2 \times 10^{-16}$ corresponds to $kT_e \sim 0.5 \text{ eV}$
- Hence conflicting requirement on ionization ($T_e > 3 \text{ eV}$) vs. vibrational excitation ($T_e < 1 \text{ eV}$)



Experiments at IPF Stuttgart

Microwave input **30 kW @ 915 MHz**



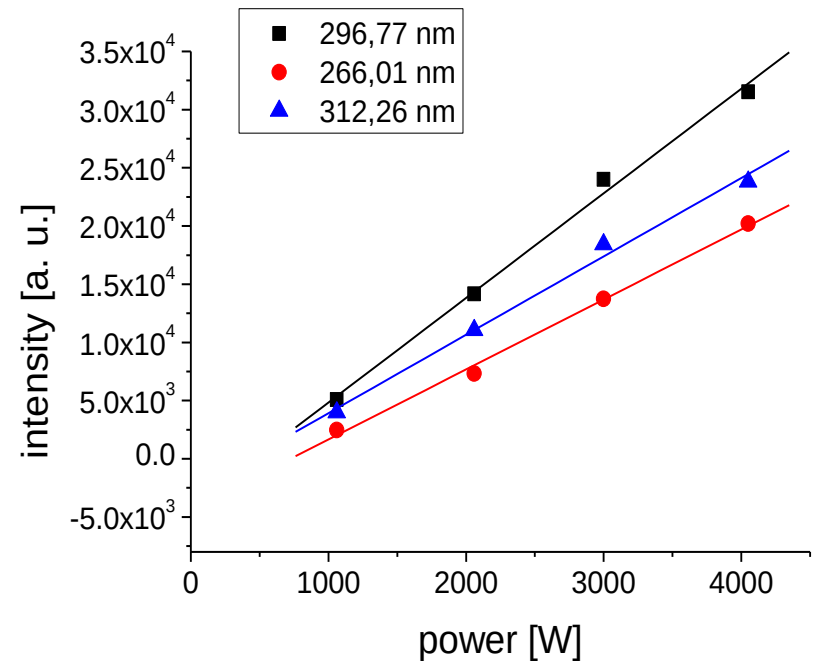
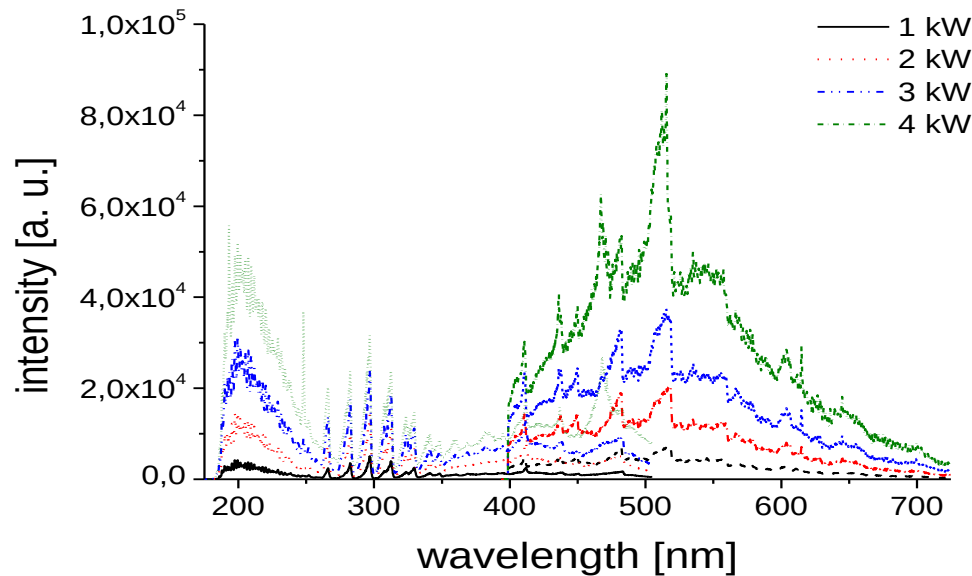
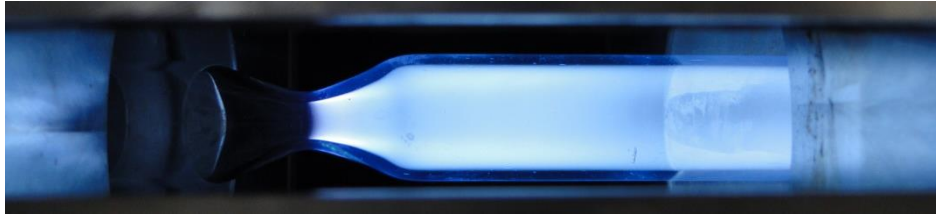
Mass
Spectrometer

Pump

High and low reduced electric field CO₂ plasma



Type I discharge, Optical emission spectroscopy



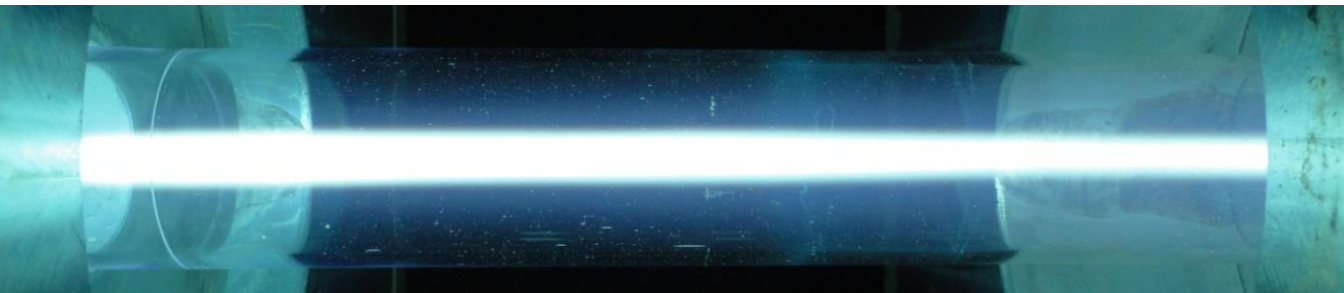
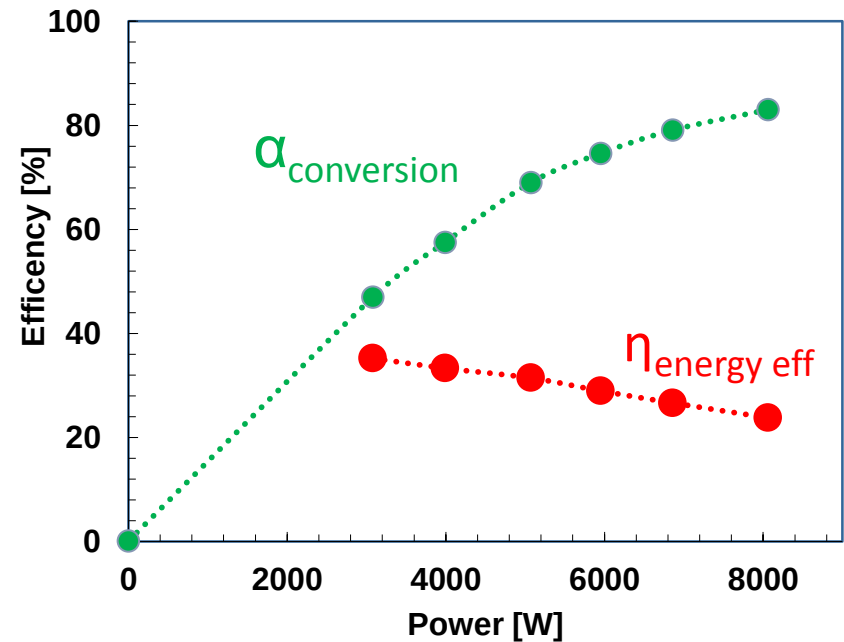
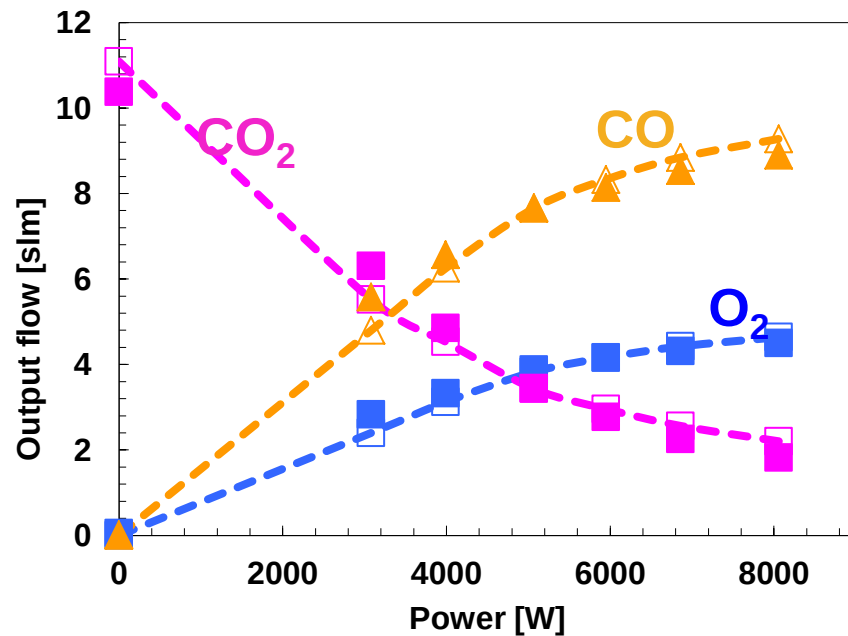
- CO third positives, fourth positives, Angstrom and triplet identified.
- CO line intensity increases linear with power



Mass spectrometry

CO produced at expense of CO₂

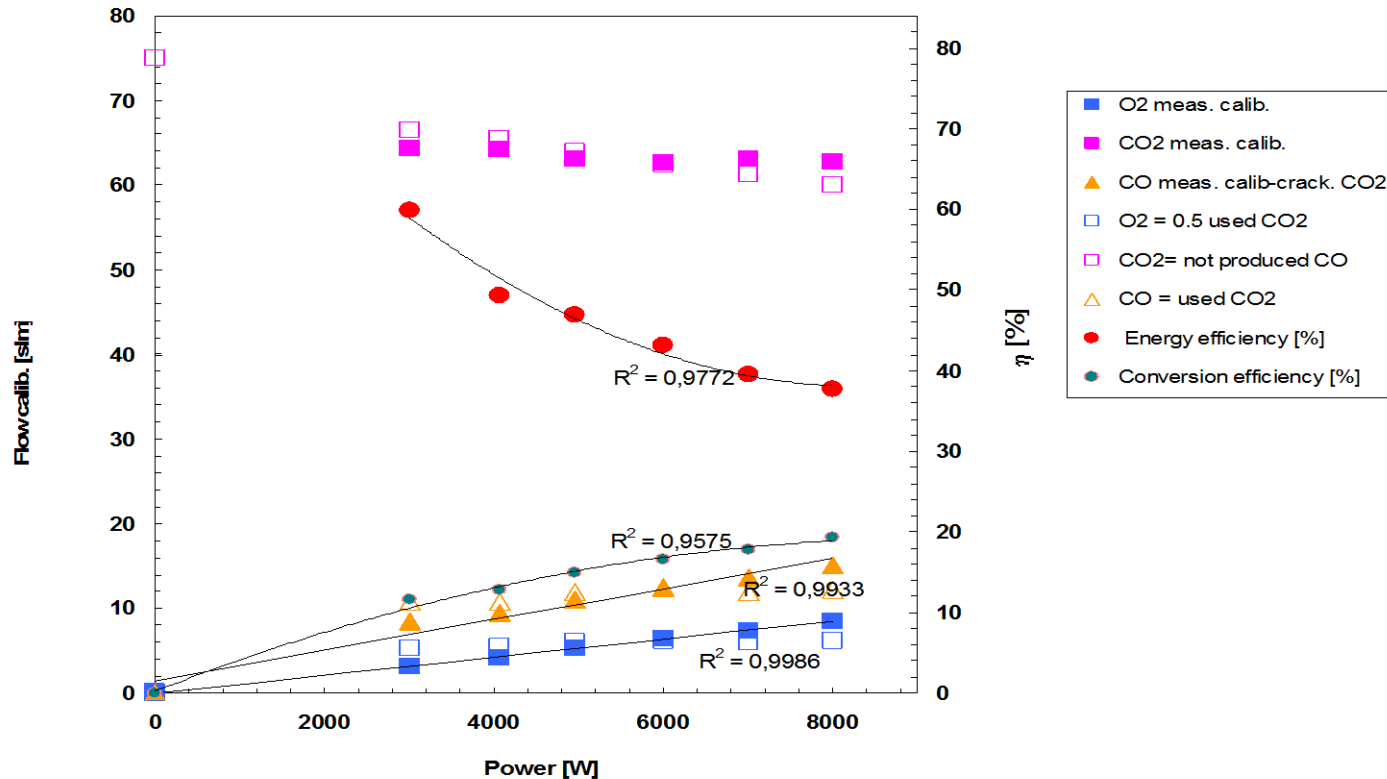
CO₂ flow 11.1 slm



$$\eta = \Delta H / E_{\text{CO}}$$

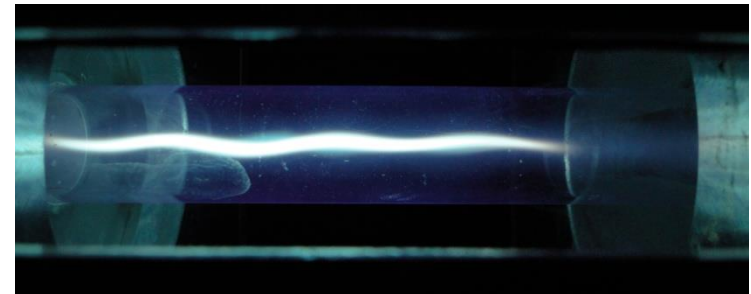


Lower reduced E field to enhance efficiency



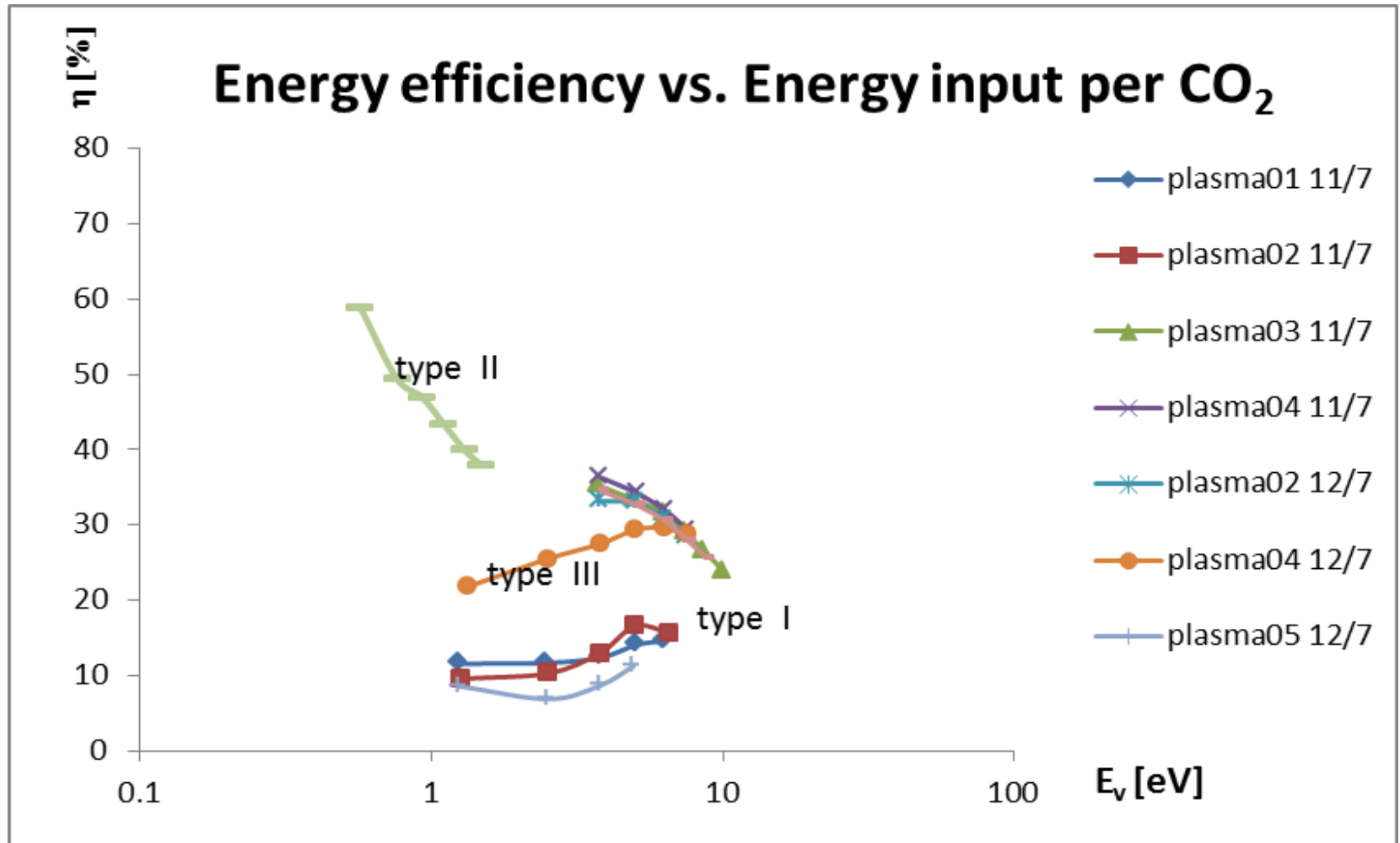
- 10 mm symmetric nozzle at exit of RF cavity
- RF input power 3020-8010 W
- Gas pressure reaction chamber 190-250 mbar
- Expansion chamber 0.3..0.4 mbar
- Gas flow 75 slm CO₂
- **Energy spent per CO₂ molecule 0.56..1.49 eV**

Gas flow →



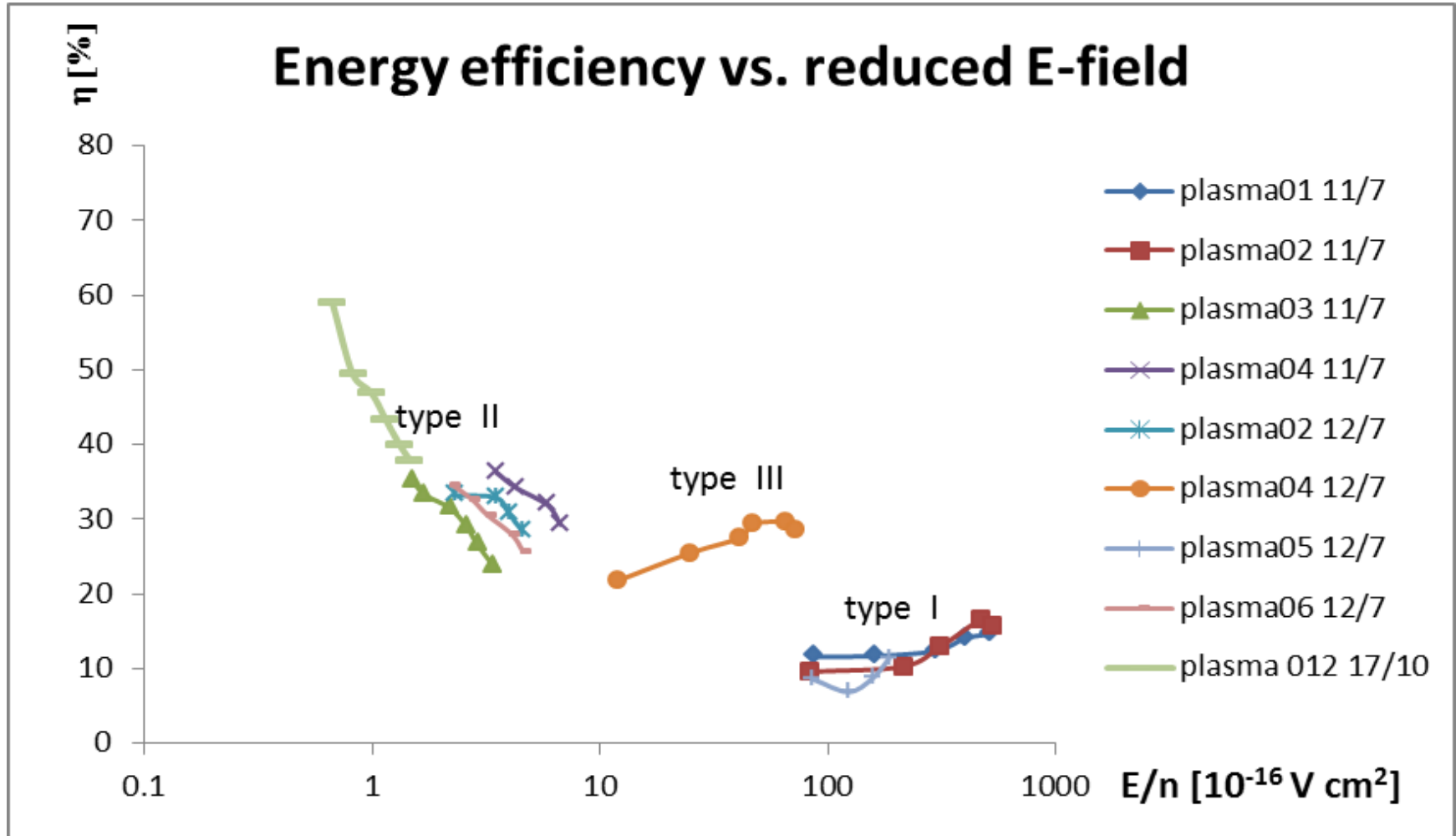


Overview of experimental results



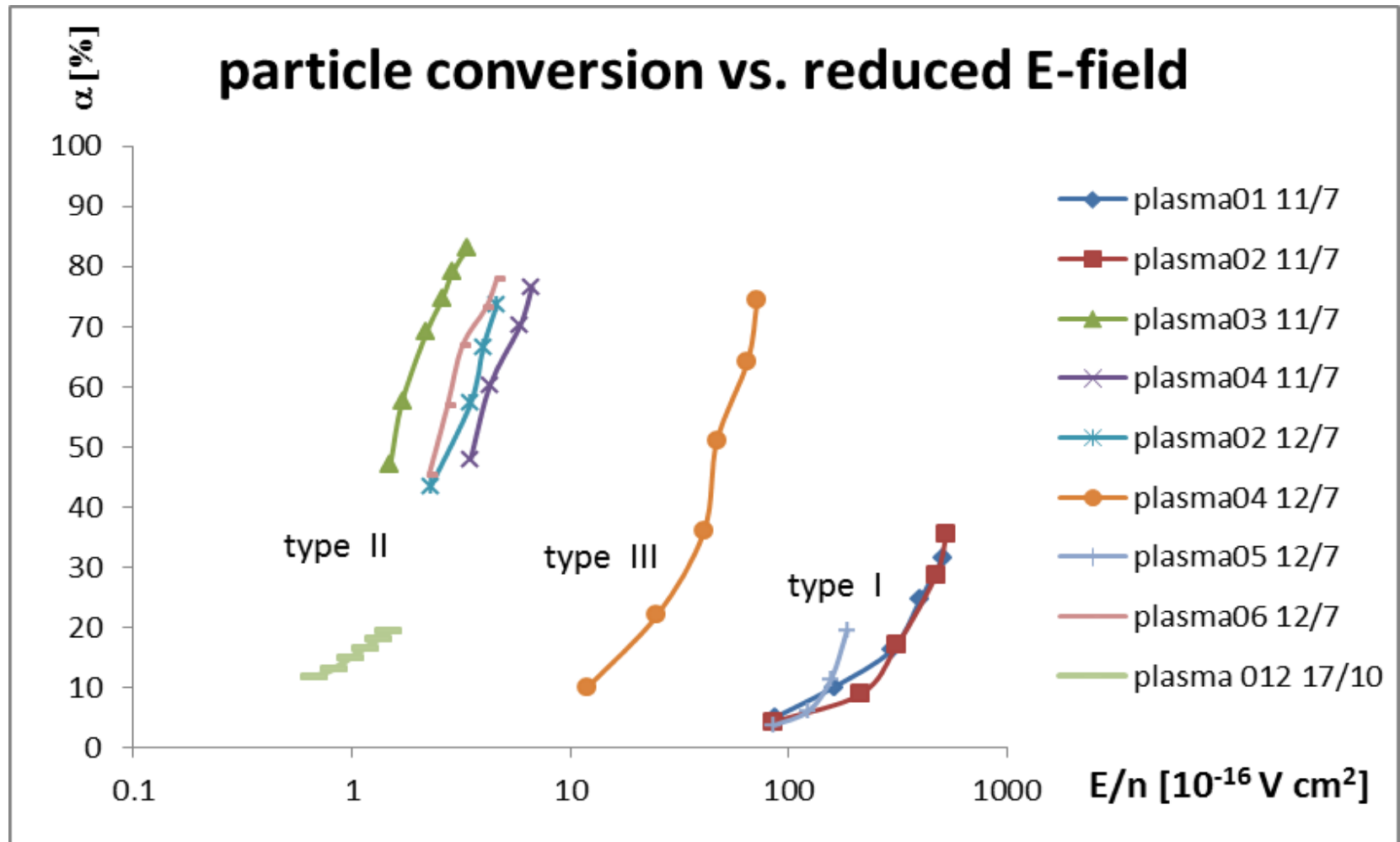


Overview of experimental results





Overview of experimental results





Conclusion

Energy efficiency >50% obtained for 10% CO₂ conversion through low temperature plasma activation.

Next Physics steps:

- **Diagnose the plasma: determine reduced electric field (incl. n_e , T_e , n_0 , T_0) and vibration state CO₂ (FTIR, Thomson scatter, CARS,..)**
- **Create super cooled gas stream $T_0 \sim 100\text{K}$ to lower vibration de-excitation**
- **Separate out flow CO from CO₂ (low conversion factor)**

Next System steps:

- **DIFFER experimental facility (1.3 kW 2.45 GHz) built**
- **100 kW 915 MHz DIFFER facility planned**
- **Design and development of output gas separation system**



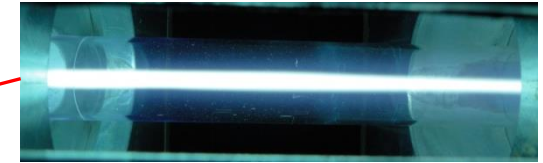
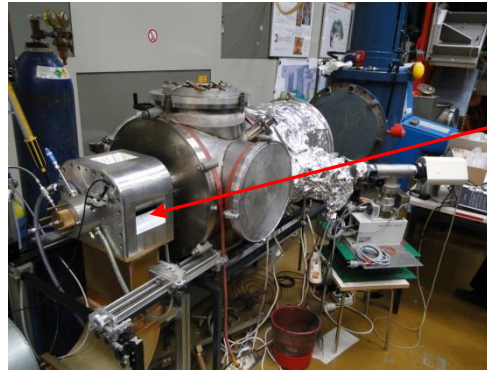
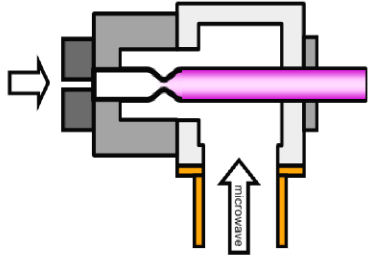
Outlook

- **System approach, including**
 - carbon capture serving as input
 - Separation of CO and H₂ at output
 - processing to gaseous (S) or liquid fuel (F-T)
 - System engineering
 - Economic and social assessment



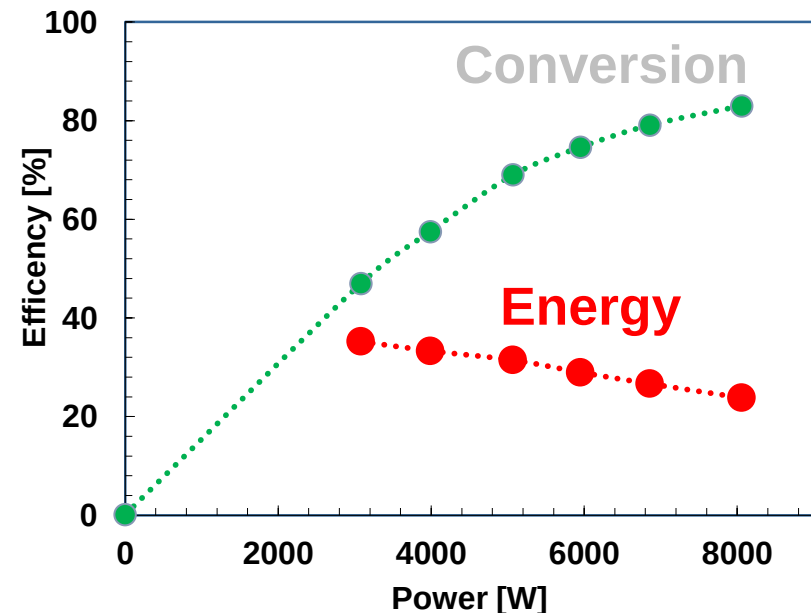
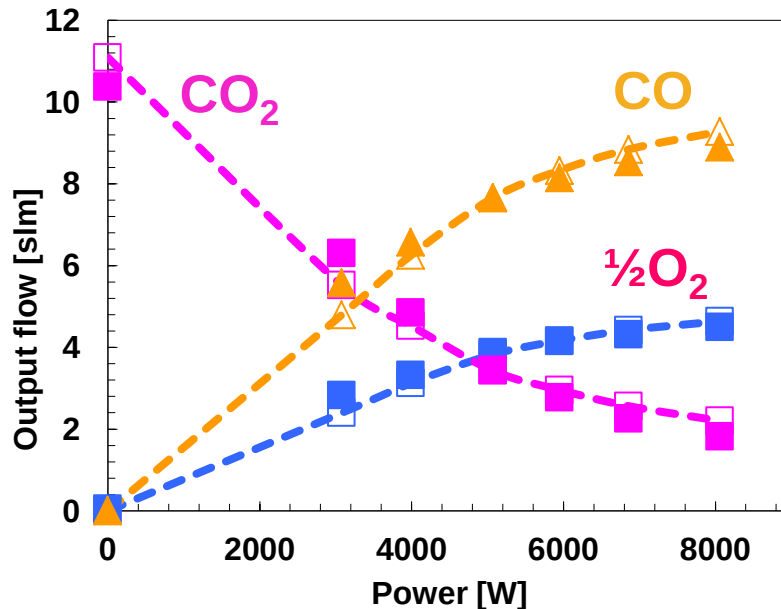


Flying start by DIFFER & IPF Cooperation



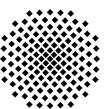
**Plasma by
915 MHz microwaves
max 30 kW**

✓ **CO produced at expense of CO₂**



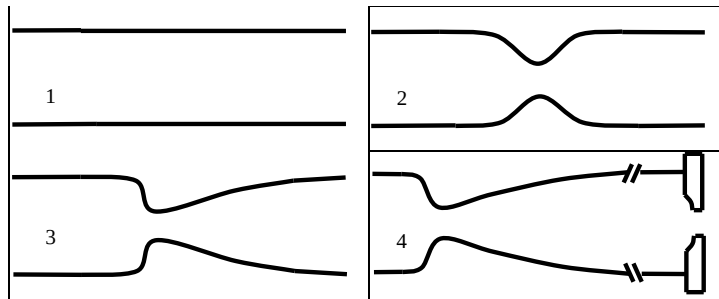
✓ **Measurements (solid) by mass spectrometry**

Constant CO₂ flow of 11.1 SLM (2.3 kW @ 100% Energy efficiency)



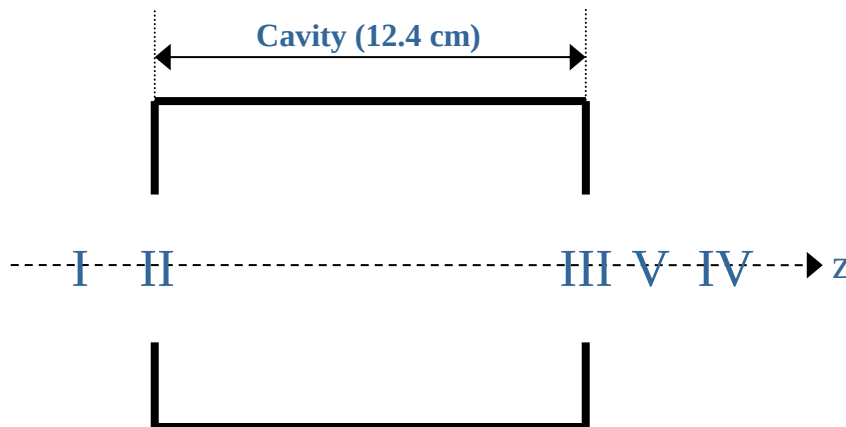


Various nozzle positions/types explored



Nozzle configurations/types:

- 1. Straight ($\varnothing$ 26 mm)
- 2. Symmetric ($\varnothing$ 26 mm with $\varnothing$ 5 mm and $\varnothing$ 10 mm throat)
- 3. Laval ($\varnothing$ 26 mm with $\varnothing$ 5 mm throat)
- 4. Double nozzle (Laval combined with copper exit nozzle of $\varnothing$ 5 mm)

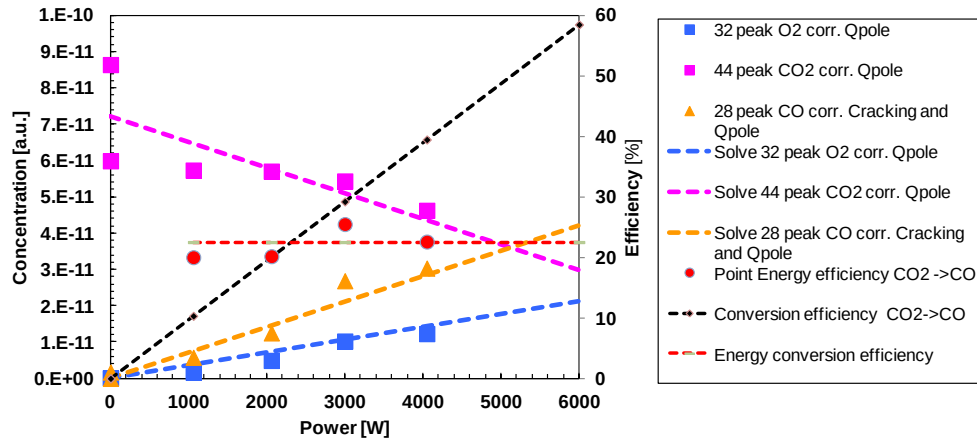
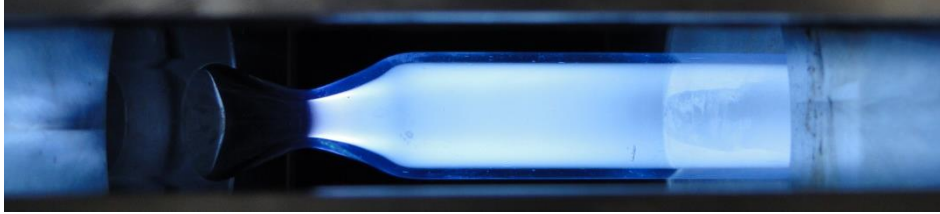


Nozzle Positions

- I $z = -3$ cm
- II $z = 0$ cm
- III $z = 12.4 + 0$ cm
- IV $z = 12.4 + 5$ cm
- V $z = 12.4 + 3$ cm



Particle conversion α and energy efficiency η



- $\alpha = \text{CO}_{\text{out}}/\text{CO}_{2\text{in}} = \text{CO}/(\text{CO}+\text{CO}_2)$
- $W = NF/60 \times 22.4 \text{ (eEv)} \text{ [eV/s]} = C \text{ (Ev/H)} F \text{ [W]} \text{ [1]}$
- $I_{\text{co}} = aW$; $I_{\text{co2}} = -aW + b$ and $I_{\text{o2}} = \frac{1}{2} aW$, yielding: [2]
- $\alpha = a/b W$ at fixed F [3]

$$n_E = H/E_{\text{co}} = H/(E_v/a) = a H/E_v = [\text{Eq.3}] = a/b W H/E_v = [\text{Eq.1}] = a/b CF$$

$$E_v = (a/n_E) H = 2.9 (a/n_E) \text{ [eV]}$$



Electron energy loss in CO₂ plasma

electrons



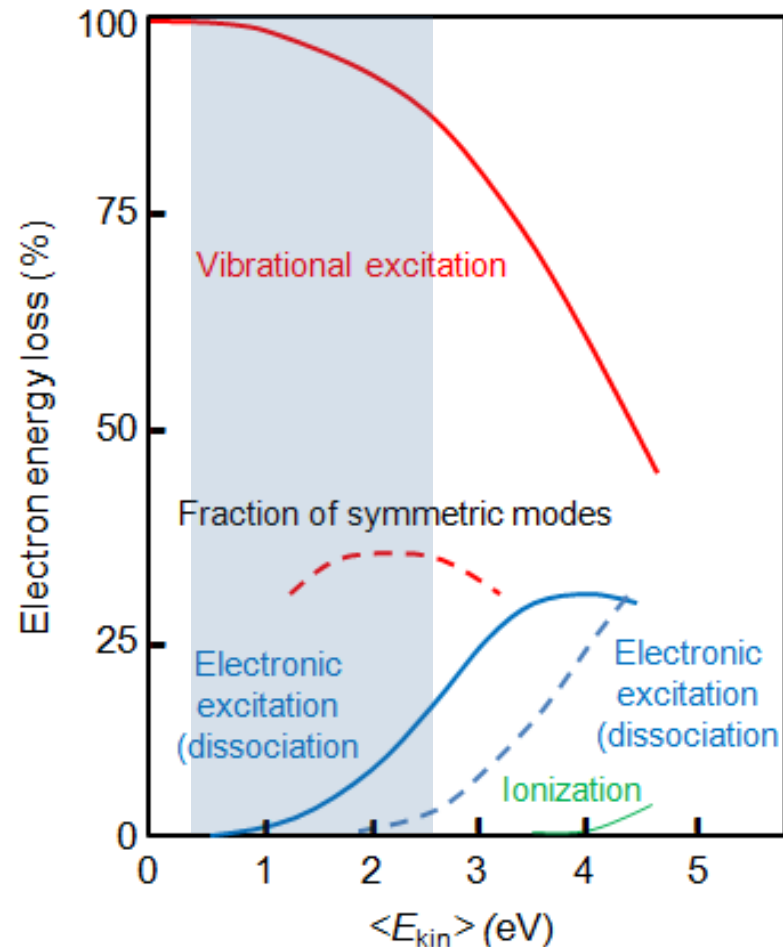
electric field



collisions



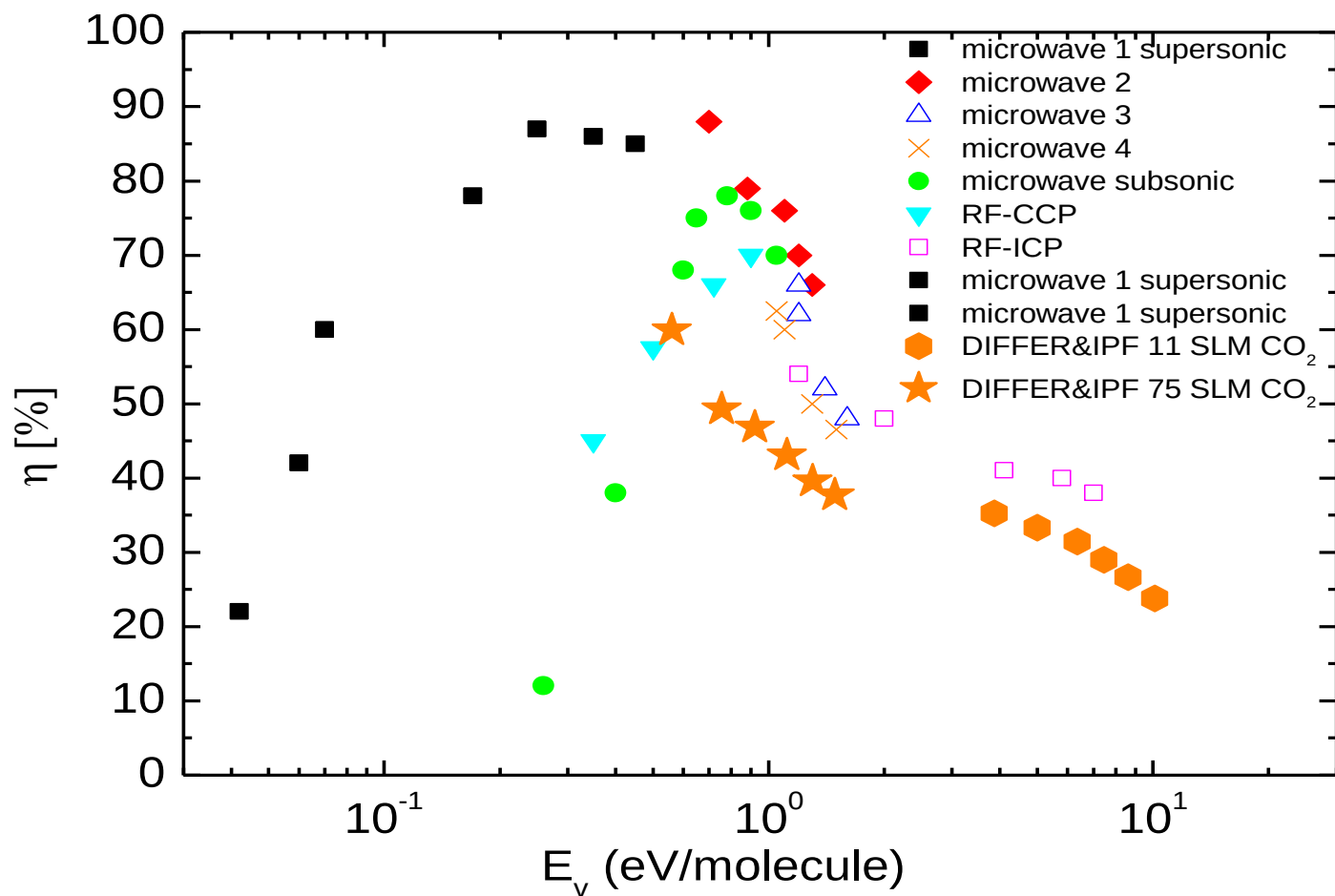
vibrational
excitation



Electron energy loss depends on reduced electric field $\rightarrow$ depends on average electron energy



Energy per incoming CO₂ molecule to be around 0.4 eV

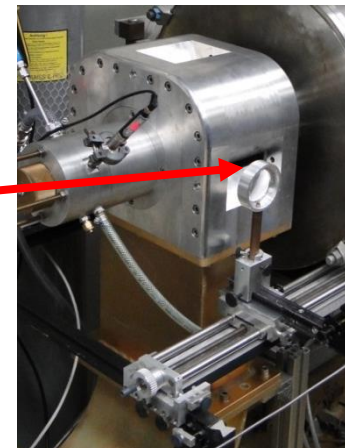
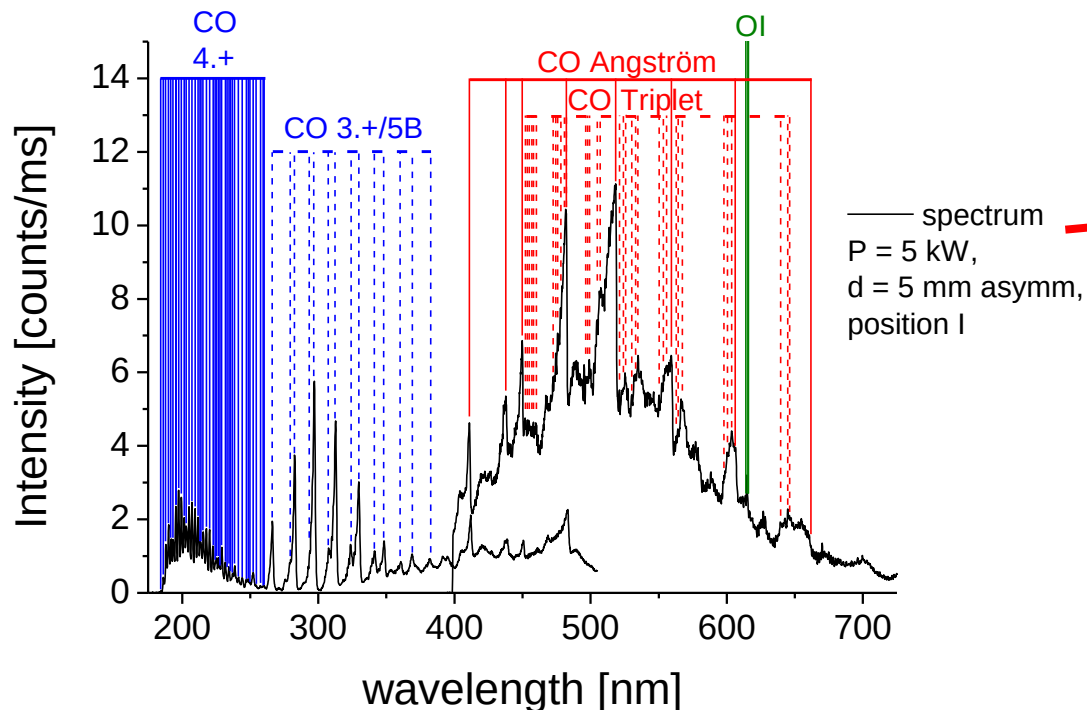


Russian Experimental Results and recent DIFFER results



Optical Emission Spectroscopy

- electronic transitions CO and C2 observed
- Insight in vibrational excitation CO (and CO₂)



Expanding plasma at 0.8 mbar plasma pressure, high reduced E field

CO **4th positive** system, singlet transition $A^1\Pi > X^1\Sigma$

CO **3th positive** system, triplet transition $b^3\Sigma^+ \rightarrow a^3\Pi$

CO **Ångström** $B^1S^+ \rightarrow A^1P$

C2 Swan band of C2 $A^3\Pi_g > X^3\Pi_u$

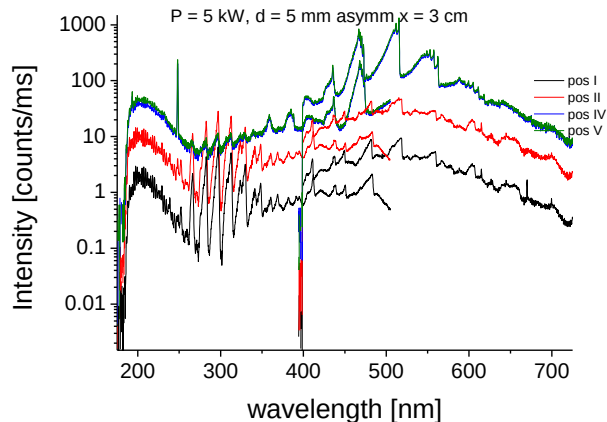
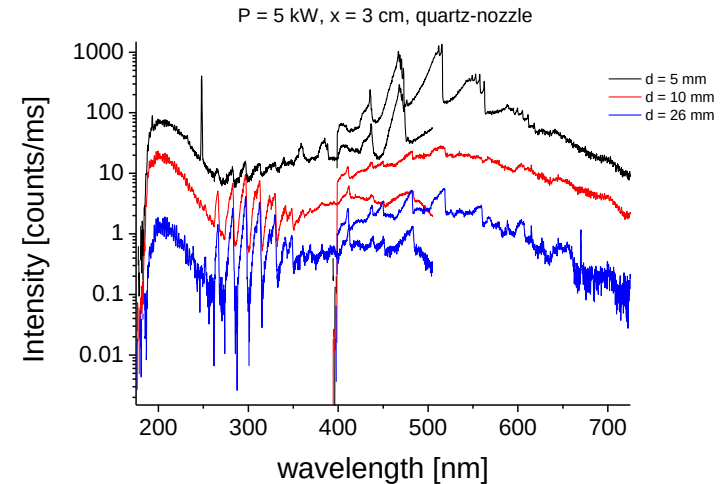
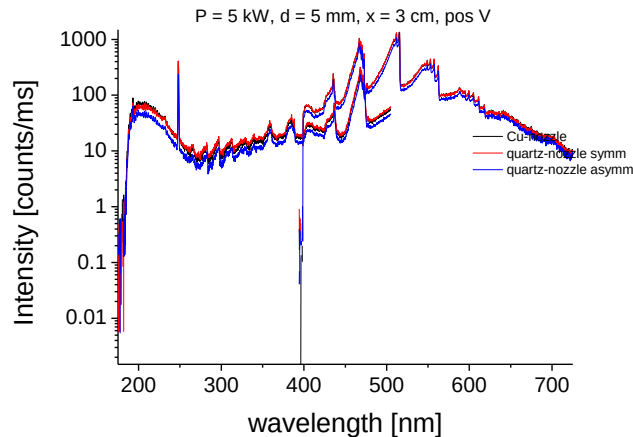
CO **Triplet** System $d^3\Delta > a^3\Pi$

CO₂⁻, CO₂⁺⁻, and CO⁺⁻-bands as well as atomic carbon lines are not observed

[Pearse and Gaydon]



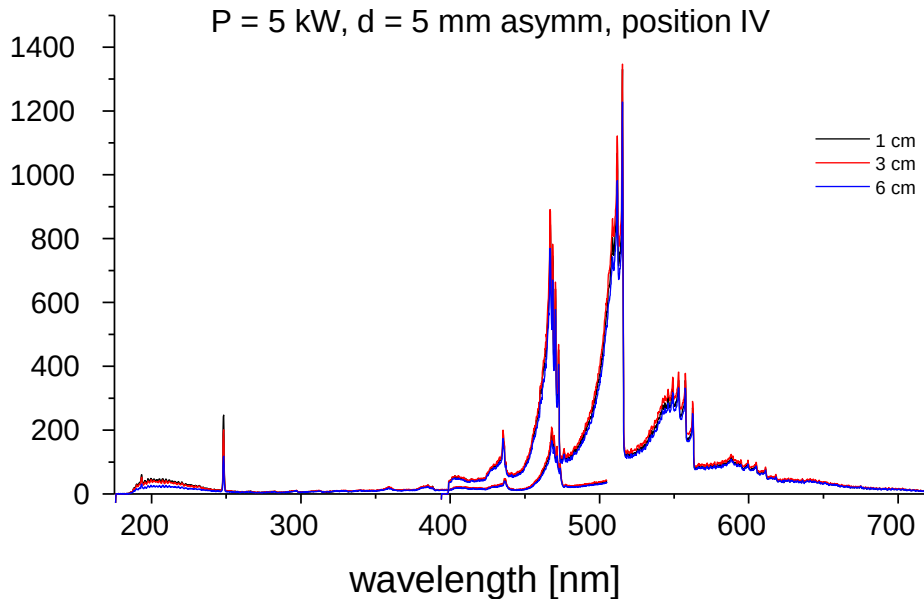
Overview CO Spectra in UV and visible



- Nozzle shape: hardly any difference observed
- Nozzle dia. change: transition from type I to type II discharge, intensity x 100
- Nozzle position at entrance or exit, dramatic change in efficiency observed

Spectra type II discharge

Intensity [counts/ms]



- Spectra dominated by Swan C2 band
- Some atomic Oxygen and atomic carbon
- Intensity factor 100 increase

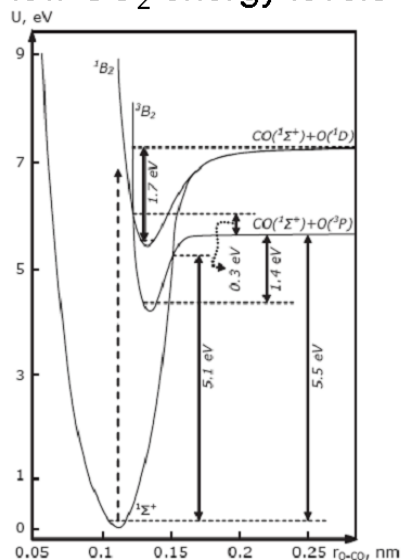
C2 Swan band $A^3\Pi_g > X^3\Pi_u$

λ [nm]	438.3 (2.0)	473.7 (1.0)	516.5 (0,0)	563.6 (0.1)	619.1 (0.2).
I [au]	2	9	10	8	3
λ measured	436	468/469	516	567/568	614/617

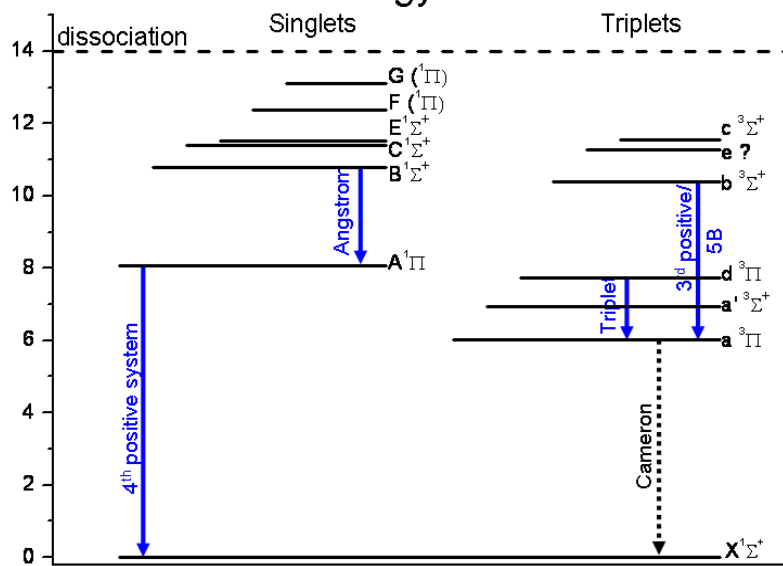


Energy diagram of CO₂ and term levels CO and O

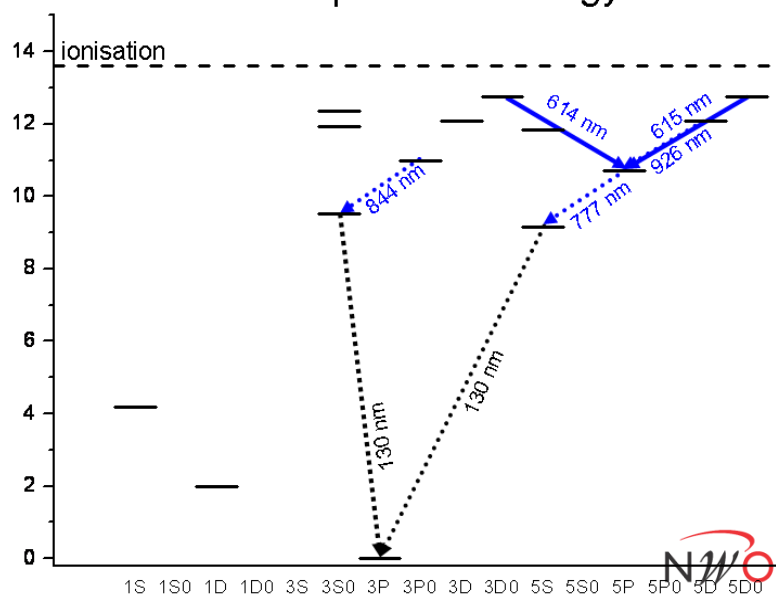
low CO₂ energy levels



CO energy levels



some important O energy levels





Identification CO spectral lines in UV-visible

λ [nm] 3 rd pos P&G pg110	283.31	297.7	313.4	330.5	349.3	369.9
λ [nm ± 2 nm] measured	281	295	311	329	347	367
I [au]	10	9	8	7	6	2

5B band CO P&G λ [nm]	I [au]	spectrum measured [nm ± 2 nm]	E [eV] calculated	ΔE [eV]
266.5 (1.0)	8	264	4.69	0.236
279.3	2	278	4.453	0.214
293	1	292	4.24	0.194
307.9	5	307	4.046	0.177
324.2	6	320	3.869	0.195
341.9	5	337	3.674	
361.27	5	367		
382.51	2	381		

λ [nm]	412.36	439.31	451.09	483.53	519.82	561.02	607.99	662.03
λ measured	411	437	450	482	518	560		
I [au]	7	8	10	10	10	10	9	7

λ [nm]	480.67	497.9	505.27	523.84	601.05	643.31
λ measured			506	525	603	645
I [au]	8	6	8	5	8	10



Cross sections Itikawa

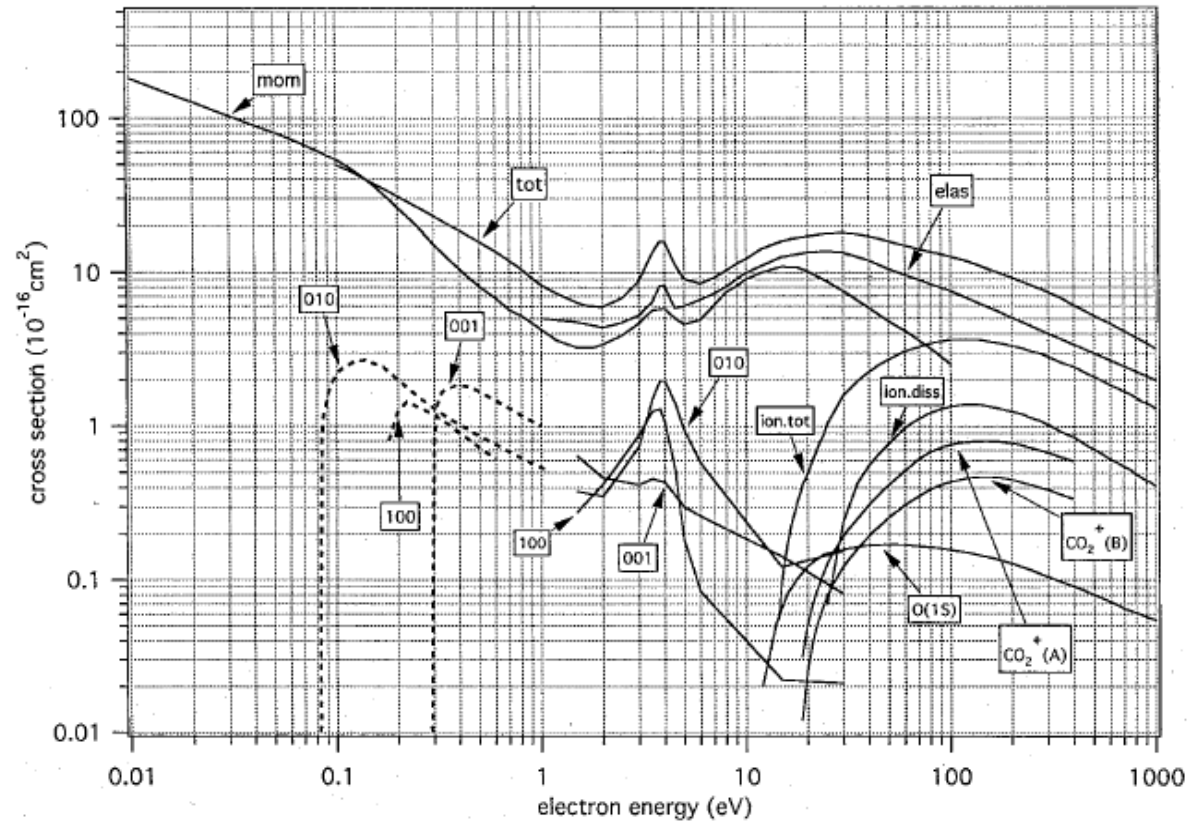


FIG. 20. Summary of the recommended electron collision cross sections for CO₂.



Sub-sonic and supersonic flow

- Theoretically, V-T relaxation decreases exponentially with gas temperature.
- Gas expansion by nozzle to lower gas temperature.
- Choked flow ($M=1$) obtained in discharge region
- Transition sub-sonic to super-sonic obtained by increase power at constant flow (gas temperature)
- For choked flow $v_F = c_s$, the relation between temperature and pressure becomes: $kT = \gamma/m_i \{pO/F\}^2$



Understanding CO₂ dissociation by plasma activation

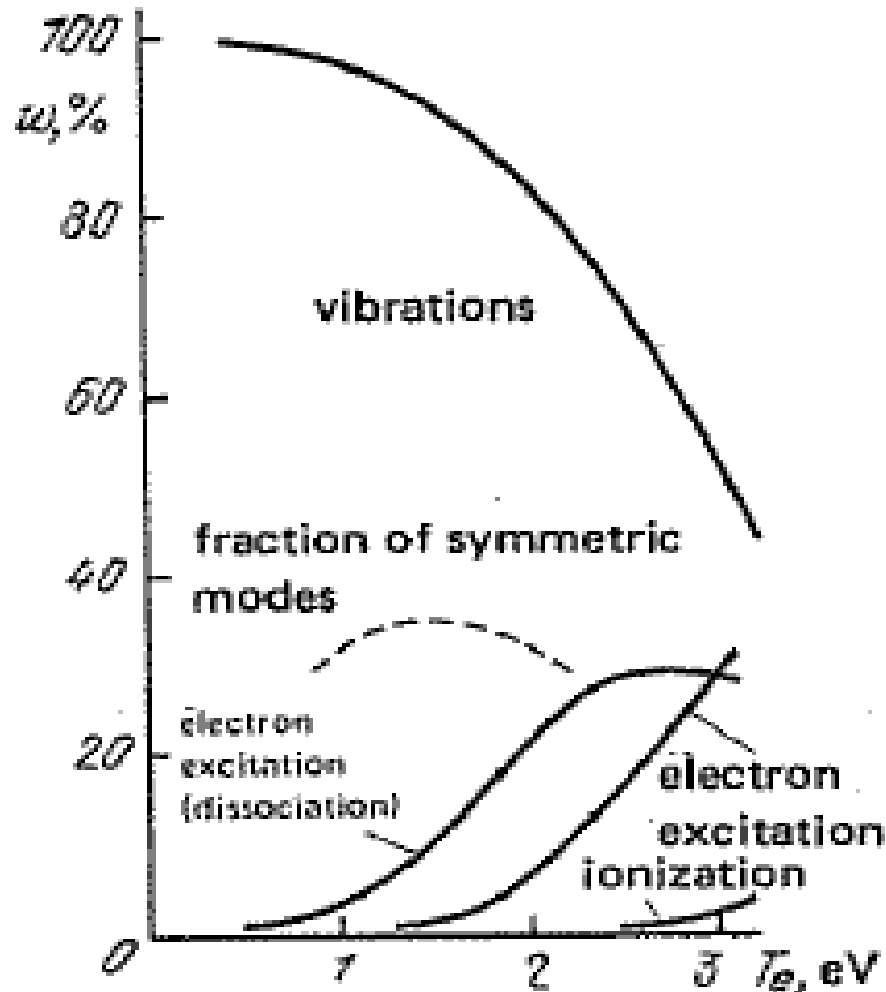
- How to make sense out of these experiments?

Answer:

- Reduced electric field $E_r = E/n$ turns out to be key
- One century old concept, yet not appreciated



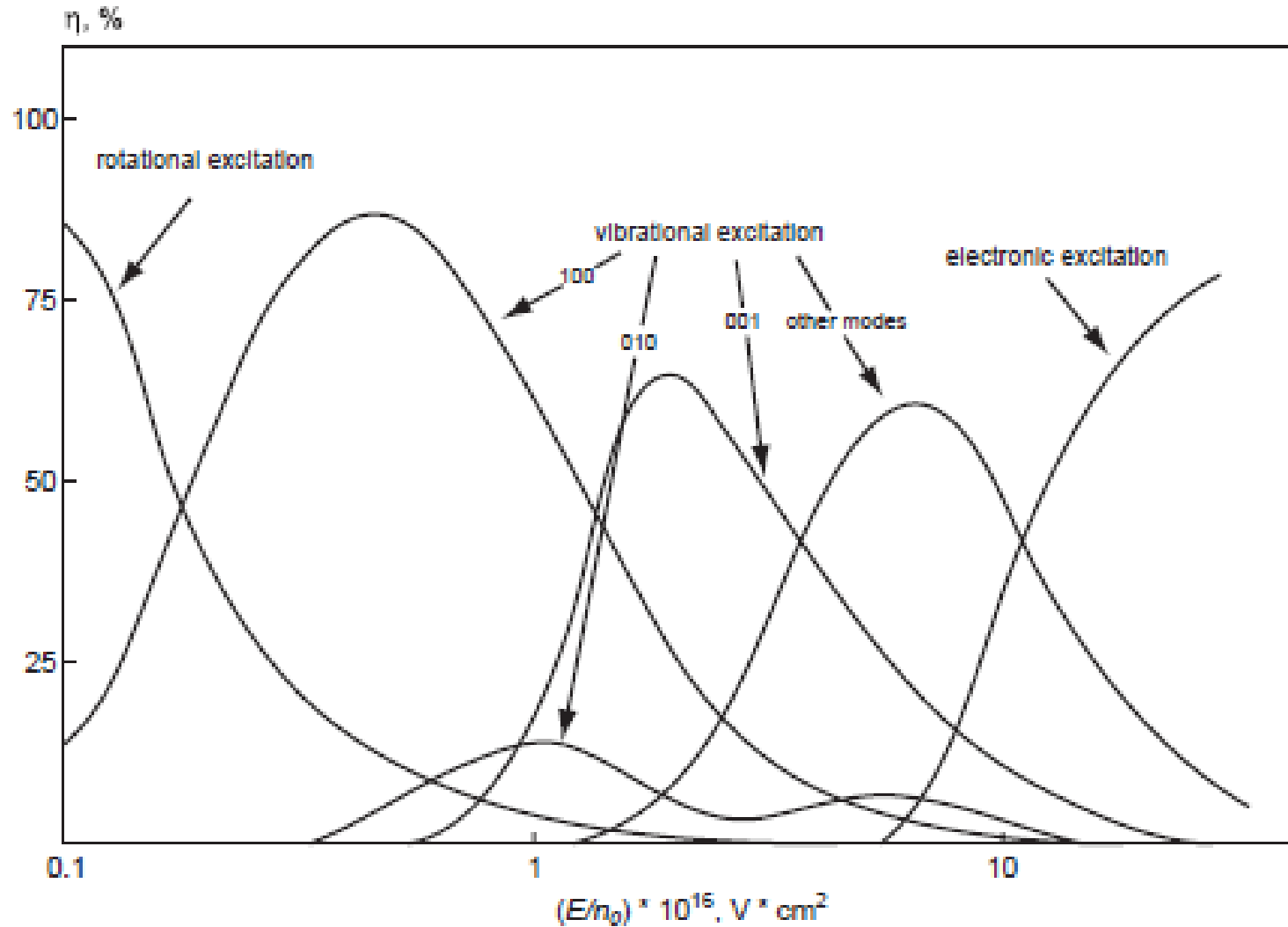
Electron excitation CO₂ vs electron energy



- vibrations excitation requires electron energy $T_e < 1 \text{ eV}$
- Ionisation starts at $T_e > 2 \text{ eV}$
- These two requirements cannot be optimally fulfilled simultaneously



Electron excitation CO₂ vs reduced E field



ν_3 (001) asymmetric stretch mode @ $E/n=2 \times 10^{-16} \text{ Vcm}^2$



50% mixture CO₂-CO

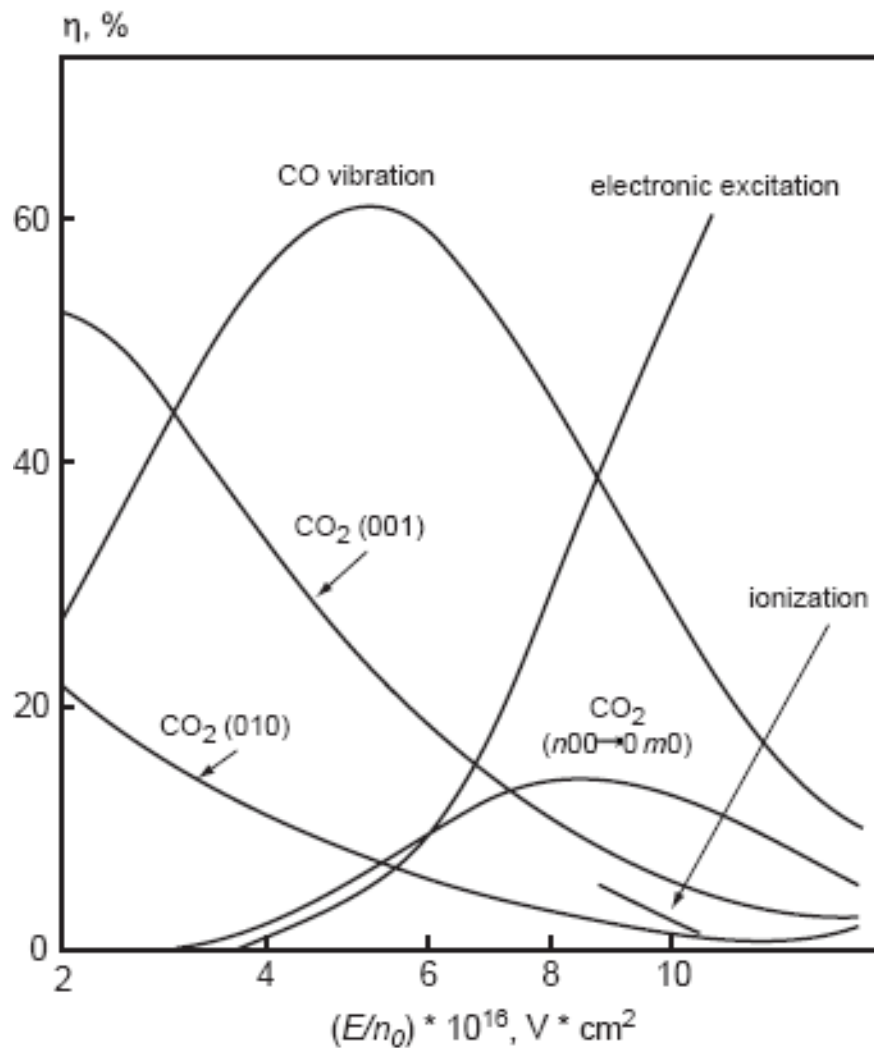


Figure 2-33. Electron energy distribution between different excitation and ionization channels in CO₂(50%)–CO(50%) mixture.



Characteristic time scales and lengths

- typical electric field strength on axis RF cavity $E=27\text{kV/m}$ @ 915MHz ($P_{\text{RF}}=3\text{kW}$)
Half cycle time $\tau=0.546\text{ ns}$
- distance travelled by an electron in vacuum accelerated in this E-field during a half cycle $s=\frac{1}{2}(eE/m)t^2 \sim 720\text{ }\mu\text{m}$
- In contrast, CO₂ ions move only 9 nm, remaining essentially stationary
- The speed acquired by electrons in this half cycle $v=at$ becomes $v= 2.6 \cdot 10^6\text{ m/s}$, i.e. an energy of $\sim 20\text{eV}$
- At this energy De Broglie wavelength $\lambda=h/p$ is 0.28 nm, hence of the size of the CO₂ diameter, i.e. ineffective in vibrational excitation
- The electron neutral collision times follow from the cross section data Itikawa
- At low pressure (1 mbar) $\lambda_{\text{tot}} \sim 370\text{ }\mu\text{m}$, i.e. $\lambda_{\text{tot}} \sim s$ and electron gains energy
- At high pressure (100 mbar) $\lambda_{\text{tot}} \sim 4\text{ }\mu\text{m}$, i.e. at 1/100 of RF cycle electron is knocked off-course, not able to gain speed
- interaction time electron - CO₂ molecule $\tau_{\text{exc}} = 2a_0/v_e = 0.39\text{ fs}$ for a 1eV electron, much faster than vibration time $\tau_{\text{vib}} = 14\text{ fs}$, ie CO₂ frozen during interaction
- CO₂ - CO₂ collision time $\tau_{\text{int}} = 4a_0/v_{\text{th}} = 1.2\text{ ps}$ ie molecules locked during 100 oscillations



Electron - CO₂ interaction times

Table II Characteristic time scales and dimensions of the CO₂ plasma

p mbar	τ_{exc} fs	τ_{vib} fs	τ_{etot} ps	τ_{ev} ps	τ_{nn} ns	λ_{nn} μm	λ_{ev} μm	λ_{tot} μm	s μm
1	0.39	14 @	62	12E3	130	50	74E3	370	720 @
100		$\lambda=4.2\mu\text{m}$	6	62	1.3	0.5	37	3.7	E=27kV/m

$$\tau_{\text{exc}} \ll \tau_{\text{vib}} \ll \tau_{\text{int}} \ll \tau_{\text{etot}} \ll \tau_{\text{ev}} \ll \tau_{\text{nn}} \sim \tau_{\text{RF}} @ 100\text{mbar}$$

CO₂ molecule is vibration excited in the ground state $^1\Sigma^+$ by high rate electron impact

VV exchange process populates the higher 1B_2 vibration energy levels of the CO₂ molecule



Plasma parameters

- In general plasma collision times much longer than neutral collision times. However, at extremely low electron temperature and high density prevailing in our experimental situation, these times become near neutral collision times. Hence, Coulomb collisions play some role in thermalisation.
- $T_{ee} < T_{ei} \ll T_{ii} < T_{ie}$

n_i	$T_e=0.1 \text{ eV}$	$T_e= 1 \text{ eV}$	$T_e=10\text{eV}$	$T_e=100\text{eV}$
10^{18} m^{-3}	1.3 ns	32 ns	0.84 μs	23 μs
10^{19} m^{-3}	0.15 ns	3.6 ns	0.092 μs	2.5 μs



Debye length and number of particles Debye sphere

n_e	0.1 eV	1 eV	10 eV
10^{18} m^{-3}	$2.35 \text{ } \mu\text{m}$	$7.45 \text{ } \mu\text{m}$	$23.5 \text{ } \mu\text{m}$
10^{19} m^{-3}	$0.75 \text{ } \mu\text{m}$	$2.35 \text{ } \mu\text{m}$	$7.45 \text{ } \mu\text{m}$

- Debye length small compared with plasma dimension
- Number of particles in Debye sphere is unusually small

N_{λ_D}	0.1 eV	1 eV
10^{18} m^{-3}	54	1720
10^{19} m^{-3}	17	550



V-T Relaxation efficiency is function ionisation degree

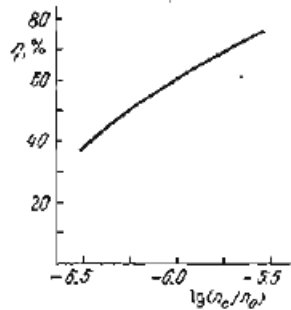


FIG. 10. Relaxation efficiency of CO₂ dissociation vs degree of ionization ($E_v = 0.5$ eV/molecule).

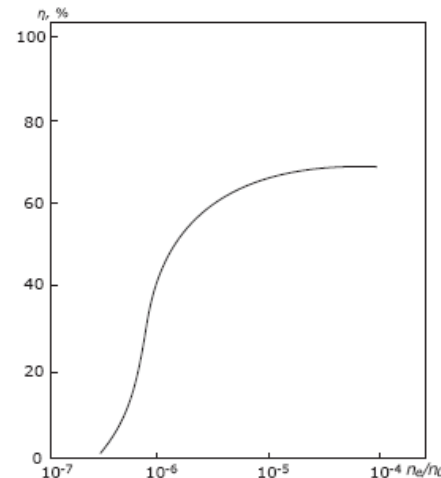


Figure 5-17. Energy efficiency of CO₂ dissociation in plasma at the specific energy input $E_v = 0.5$ eV/mol as a function of plasma ionization degree n_e/n_0 .

η_{VT} is the fraction of vibration energy going into dissociation over the fraction going into translation energy. This η_{VT} turns out to be a function of ionisation degree.

Here comes the plasma!

At a pressure of 200mbar ($n_0 = 5.38 \times 10^{18} \text{ cm}^{-3}$), a relaxation efficiency $\eta_{VT} = 80\%$ is reached at an electron density $n_e = 1.7 \times 10^{13} \text{ cm}^{-3}$, i.e. quite high!



Plasma frequency as function of density

n_e	$N = 8.967 \sqrt{n_e}$
10^{18} m^{-3}	$\sim 9 \text{ GHz}$
10^{19} m^{-3}	$\sim 28 \text{ GHz}$

The RF frequency (915 MHz) is lower than the plasma frequency. Hence, the wave cannot propagate. With collisions included, the wave can propagate, but the RF field is essentially stationary for the processes considered.



Vibration energy loss

$\log k(\text{cm}^3/\text{sec})$

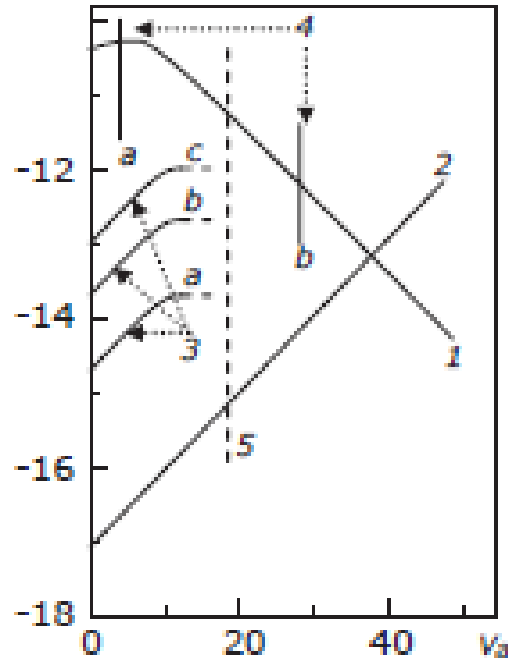


Figure 5-13. Rate coefficients of relaxation processes related to asymmetric mode of CO_2 vibrations at room temperature $T_0 = 300 \text{ K}$ as functions of number of quanta on the mode: (1) VV relaxation; (2) VT relaxation; (3) intermode VV' relaxation (assuming symmetric vibrational temperature [3a] $T_{\text{vs}} = 1000 \text{ K}$, [3b] $T_{\text{vs}} = 2000 \text{ K}$, [3c] $T_{\text{vs}} = 3000 \text{ K}$); (4) intramolecule VV' relaxation, transition to vibrational quasi-continuum (assuming [4a] equal excitation of all vibrational modes; [4b] predominant excitation of asymmetric vibrations); (5) CO_2 dissociation energy.

$$k_{\text{VT}} \approx 10^{-10} \exp \left(-72/T_0^{1/3} \right), \text{ cm}^3/\text{s},$$

- At vibration levels up to 20, the dissociation energy, the vibration pumping exceeds the translation loss



Vibration Excitation CO₂ vs. Reduced E-Field

Few electrons



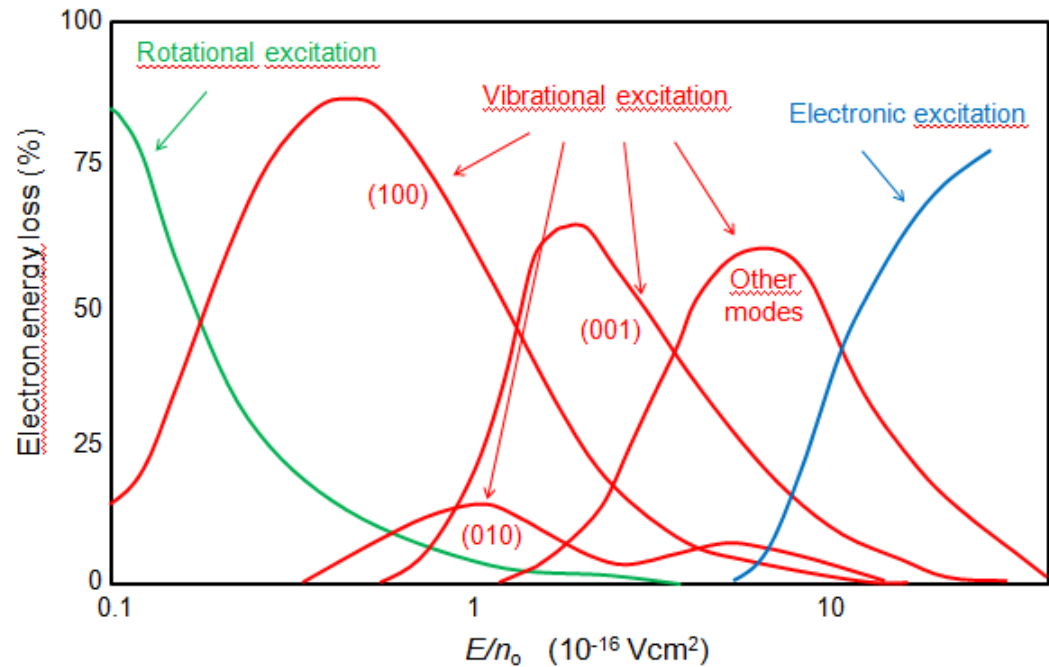
Low reduced
electric field



Electron-neutral
collisions



vibrational
excitation



ν_3 (001) = CO₂ asymmetric stretch mode



Nonequilibrium effects: Treanor distribution

electrons



electric field

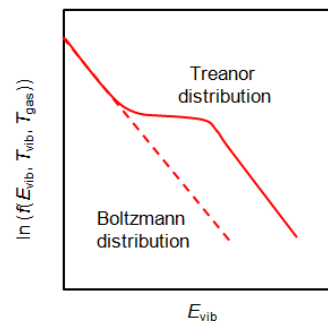
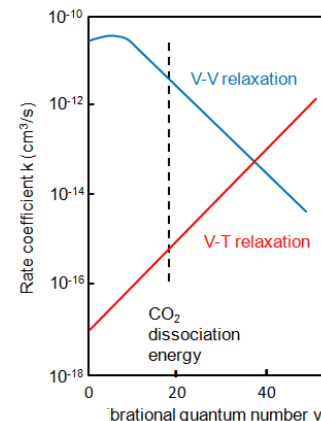
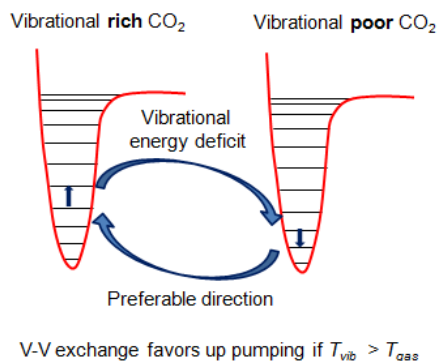


collisions



vibrational
excitation

$$T_{\text{electron}} > T_{\text{vib}} > T_{\text{gas}}$$



**Treanor distribution:
overpopulation of higher vibrational
states leads to lower activation
energies, faster kinetics**



The Energy problem

Why Solar Fuels?

Fossil fuels, albeit finite, are still abundant in terms of world energy reserves:

- Coal reserves 300 yrs (China)
- Gas expanded from 50 to 200 yrs due to fracking shale beds
- Oil demand has peaked in rich world
 - ◆ increased efficiency in petrol/diesel engines and emergence of hybrid and electric cars
 - ◆ switch to gas to power lorries, buses, ships and domestic/industrial heating systems

What problem?



The Climate problem: IPCC 5th Assessment

Problem is the CO₂ emission from burning fossil fuel

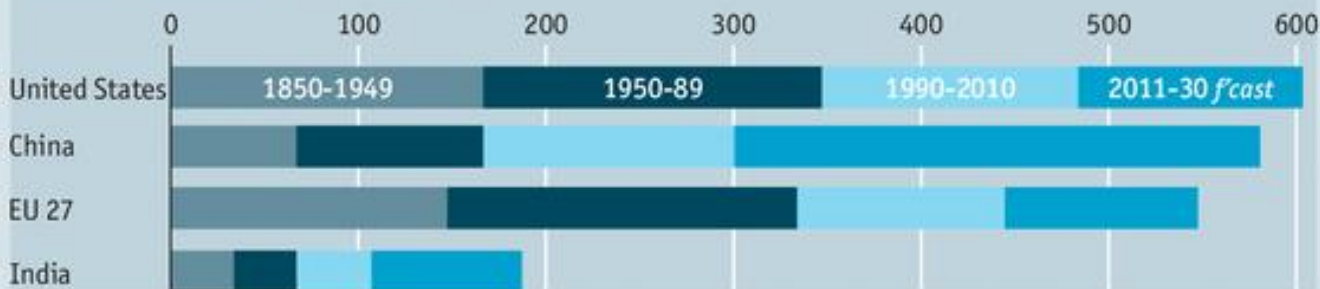
- ♦ Atmospheric concentration increased from 280ppm to 400ppm this year,
 - ♦ increased radiative forcing from 0.57 to 2.29 W/m² since industrialisation
 - ♦ Result 0.5°C global average surface temperature rise
 - ♦ + 0.5°C due to climate inertia expected later this century
- Most scientists & governments place 2°C limit on temp. rise
 - IPCC defines Carbon budget **1800 Gton CO₂** emission
(until 2011 539Gt, yearly 37 Gton global emission)
 - Budget will be spent by **2047 ±14yr** (business as usual)
 - By **2069 ±18yr** emission stabilisation (Nature 10 Oct 2013)



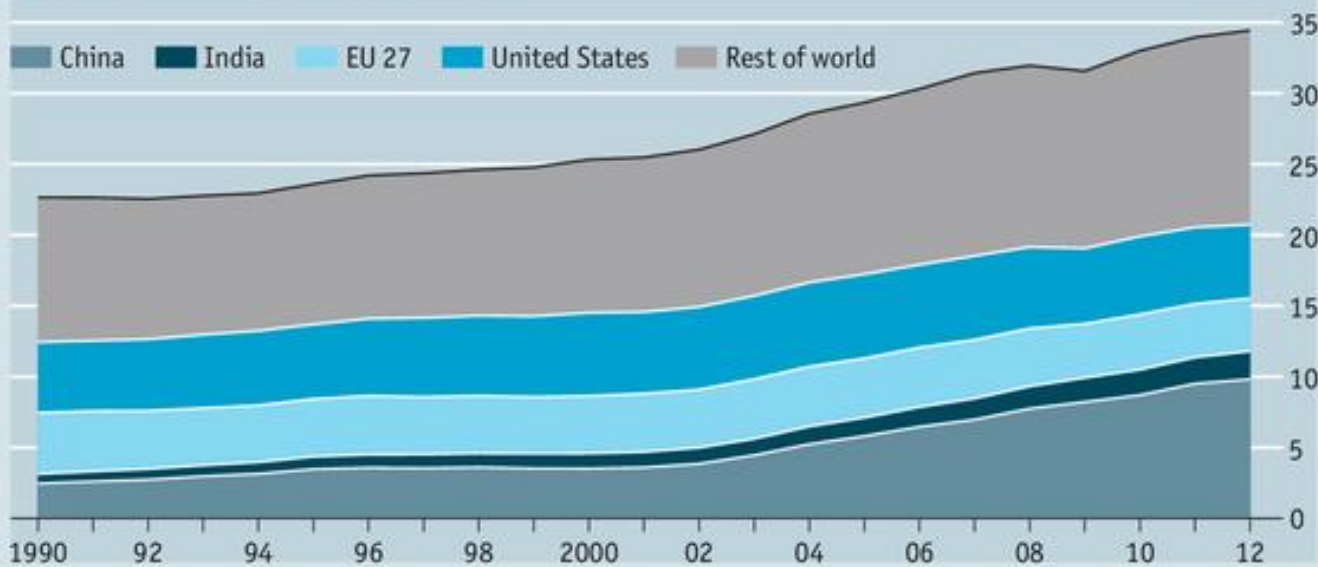
Carbon dioxide emissions past and projection 2030

Carbon-dioxide emissions

Cumulative total, gigatonnes CO₂ equivalent



Global industrial CO₂ emissions*, gigatonnes



Source: PBL Netherlands Environmental Assessment Agency

*From fossil-fuel use, cement production and transport



EU plus Aus only 13% share in CO2 emission

What happens in rich countries matters much less than it did.

Developing economies now account for over half the world's emissions, and their share will keep rising. In China, they rose less quickly last year—just 3%, rather than the 10% that has been the recent norm—and the amount of carbon emitted per unit of GDP is falling quite fast. Other nations are not making such progress. India emits much less carbon dioxide than China; but it is a bigger source than Russia, and last year its emissions rose by 7%. Soon more than half of all the carbon ever emitted by human industry and land use will have come from developing countries.

Industrial emissions of carbon dioxide continued to rise in 2012, pushing the atmospheric level ever higher, but they did so at a lower rate than in many recent years. Indeed in big developed economies they dropped—by 4% in America, where utilities were switching from coal to gas, and by 1.3% in recession-hit Europe.

Source Economist 9 November 2013



EU energy & climate policy

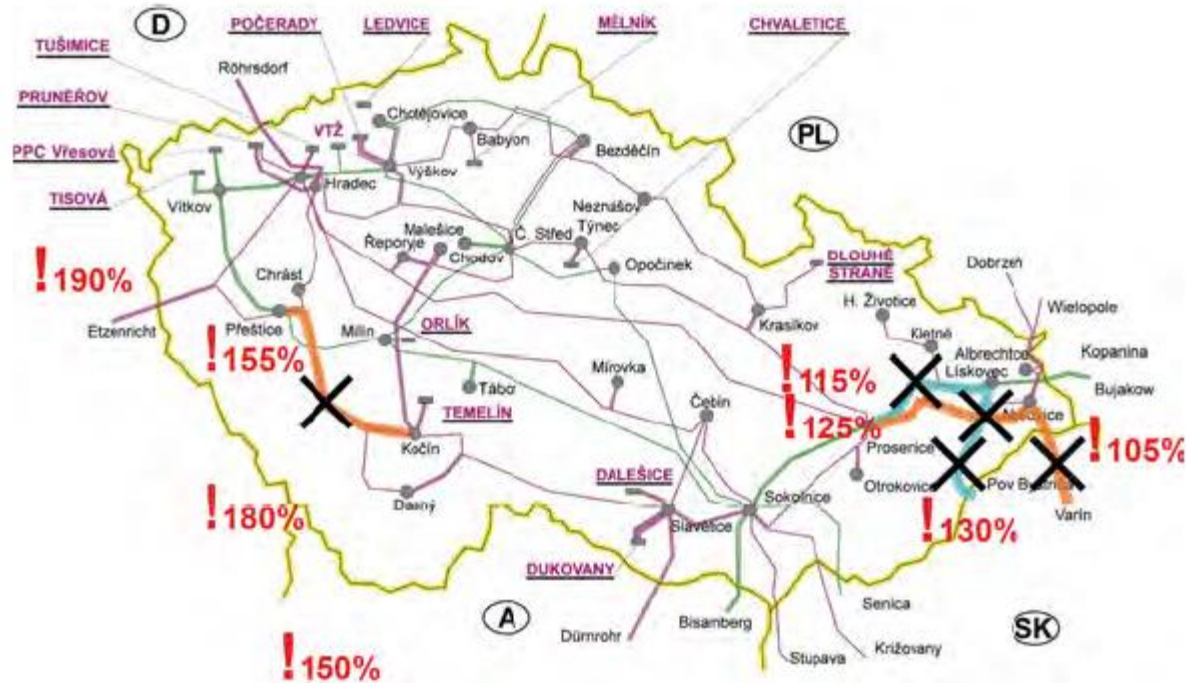
- ♦ 2020 renewable share 20%, GHG emission reduction 20%
- ♦ 2050 Roadmap GHG emission reduction 80-95%
- ♦ To date 80% fossil fuel in EU energy mix

Gas will play important role in energy transition (IEA 2011)

- CO₂ emission 80% reduced compared with coal
- Gas turbine allows rapid ramping up and down to compensate for intermittent supply by renewables
- Gas storage provides reliable and flexible means to provide back-up power to compensate for low intermittent supply
- Gas transport is factor 10 cheaper than electricity transport



Overloading of Czech electricity grid



- Black crosses indicate tripping of lines in Nov 2011-Feb 2012